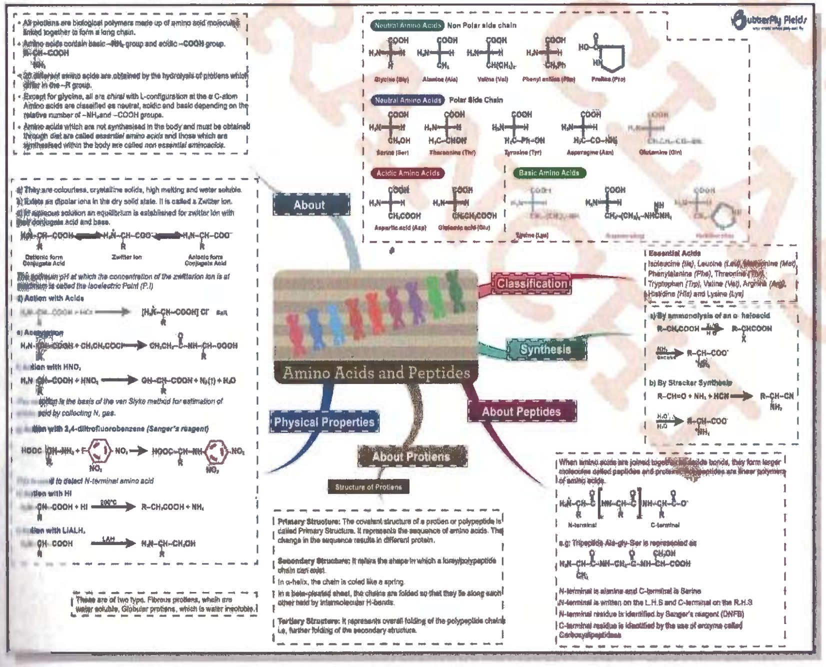
What is a Concept Map?

A Concept map is a colourful, visual form of representing key concepts covered in a chapter. It can be used as an effective overview, learning and revision tool.

At its heart is a central topic of the chapter. This is then explored by means of branches representing main ideas, which all connect to this central idea.

Concept maps are metacognitive tools that give students awareness towards the process of learning and allow the representation of knowledge structures in thematic fields.



Why use a Concept Map?

Students:

Brain is the key to effective learning for a student and more effectively the student uses it, the more successful he/she will be. If the brain of the student is stimulated with correct thinking and effective learning tools, excellent results can be obtained. **Understanding lists, lines, sequences, letters and numbers** is the work of the **LEFT BRAIN.** While **interpreting colour, images, rhythms and spatial awareness** is the work of the **RIGHT BRAIN.** Through a Concept Map both Right & Left parts of the brain would be put into use.

As a learning strategy, concept mapping stimulate learners' **commitment and interest towards the concepts,** which is very important for better learning. Students can use it as tool for effective revision of not just the current class chapters but also the topics covered previously.

Teachers:

As a teaching support, concept maps are also helpful, as they permit the establishment of networks of concepts that originate from a central concept in a coherent, structured and integrated manner, **guiding the sequence of teaching.** Concept maps can also be used by teachers for **curriculum planning.** Using this instrument means giving a new meaning to education, as well as a new meaning to the concepts of teaching, learning and assessment of learning.

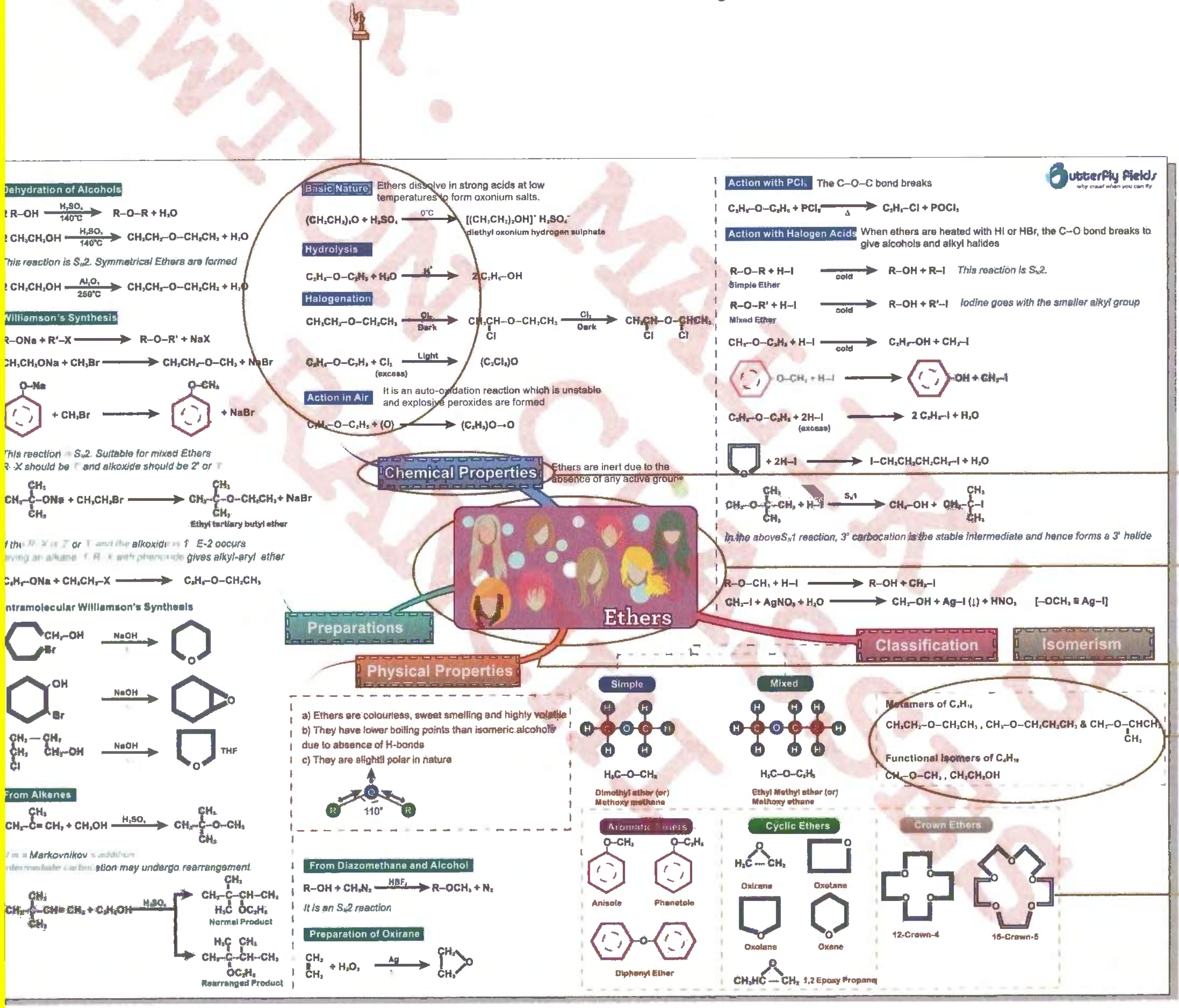
Each main topic has been transformed into a Map. For example, given below is the map for ETHERS which includes an introduction to the map.

If you take the above branch as an example, we can understand that ETHERS has a CHEMICAL PROPERTIES - Basic Nature, Hydrolysis, Halogenation, Action in Air

Important Points

- The sub-concepts and the points under them are colour coded, i.e. each branch is identified with its colour.
- Based on the colour you can identify the sub-concept under which the point will come.
- Refer to the text book and identify the first branch to start with and revise one branch at a time.

Spend some time in understanding the Maps, once you get a hang of them they are a wonderful tool to get ahead in class.



The **Sub-Concept**

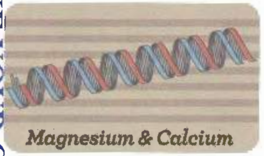
The **Central concept** of the chapter

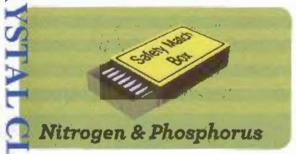
Branches – Each branch leads to a sub-concept of the chapter

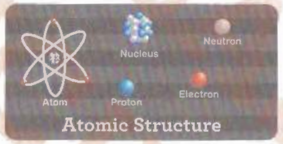
These are the important points under each sub / concept & these could be further branched out

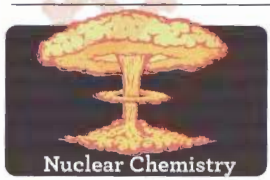
Image –Image is a mnemonic which will help you remember the concept

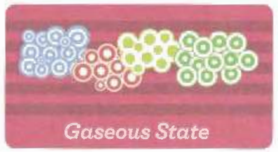
A to Z of CHEMISTRY

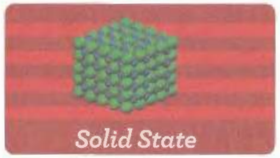


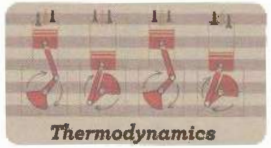


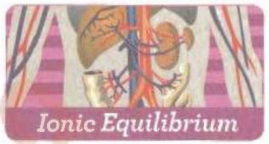


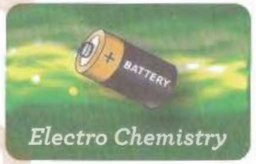




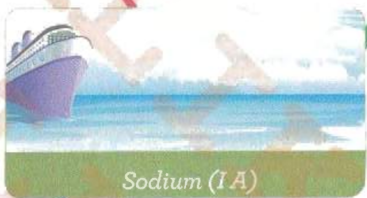








R. K. MALIK'S NEWTON CLASSES, RANCHI



General Characteristics

General Characteristics

Li Na K Rb Cs Fr - Alkali metals

ns^1 and +1 oxidation state

Soft metals and softness increases down the group with decrease in metallic bond strength

Density order: Li < K < Na < Rb < Cs
0.54 0.86 0.97 1.53 1.90 (gcm^{-1})

Atomic and Ionic radii increases down the group

$Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$
0.60 0.95 1.33 1.48 1.67 (Å)

Hydrated Ionic radii decreases down the group

$[Li(aq)]^+ > [Na(aq)]^+ > [K(aq)]^+ > [Rb(aq)]^+ > [Cs(aq)]^+$
3.40 2.76 2.32 2.28 2.27 (Å)

Ionic conductance of these hydrated ions increases down the group

$E^0_{o.p}$ value of Li is maximum due to high heat of hydration and hence it is the strongest reducing agent in solution

Li > Cs > Rb > K > Na

Flame colours	Li	Na	K	Rb	Cs
	Crimson red	Golden yellow	Pale violet	Violet	Violet

Hydroxides

NaOH - Caustic soda & KOH - Caustic potash

They are the strongest bases in solution

Highly soluble in water (Alkalis) but LiOH is least soluble

Preparation of NaOH

Causticisation *(Gossage Process)*

$Na_2CO_3 + Ca(OH)_2 \longrightarrow CaCO_3 + 2NaOH$

Lowig's Process

$Na_2CO_3 + Fe_2O_3 \longrightarrow \underset{\text{Sodium ferrite}}{2NaFeO_2} \xrightarrow{H_2O} 2NaOH + Fe_2O_3$

Nelson's Process

$NaCl\ (Solution) \rightleftharpoons Na^+ + Cl^-$

At Cathode: $Na^+ + e^- \longrightarrow Na$

$Na + 2H_2O \longrightarrow NaOH + H_2$

At Anode: $2Cl^- \longrightarrow Cl_2 + 2e^-$

Castener - Kellener process

In outer compartment

$NaCl \rightleftharpoons Na^+ + Cl^-$

At Cathode: $Na^+ + e^- \longrightarrow Na$

$Na + Hg \longrightarrow Na - Hg$

At Anode: $2Cl^- \longrightarrow Cl_2 + 2e^-$

In Middle compartment

At Anode: $Na - Hg \longrightarrow Na^+ + Hg + e^-$

At Cathode: $Na^+ + e^- \longrightarrow Na$

$Na + 2H_2O \longrightarrow NaOH + H_2$

KOH is prepared by Electrolysis of KCl solution.

Properties: Hydroxides are white solids

Deliquescent in nature

$NaOH \xrightarrow[Cl_2]{PbO_2} H_2O + Na_2PbO_3$

$NaOH \xrightarrow[Cl_2]{cold} H_2O + NaClO + NaCl$

$NaOH \xrightarrow{hot\ Cl_2} H_2O + NaClO_3 + NaCl$

$NaOH \xrightarrow{P_4} PH_3 + NaH_2PO_2$

$NaOH \xrightarrow{S} H_2O + Na_2S + Na_2S_2O_3$

$NaOH \xrightarrow{B} H_2 + Na_3BO_3$

$NaOH \xrightarrow{ZnSO_4} Zn(OH)_2 \longrightarrow Na_2ZnO_2$

$NaOH \xrightarrow{AlCl_3} Al(OH)_3 \longrightarrow NaAlO_2$

$NaOH \xrightarrow{SnCl_2} Sn(OH)_2 \longrightarrow Na_2SnO_2$

$NaOH \xrightarrow{AgNO_3} Ag(OH) \longrightarrow \underset{Brown}{Ag_2O}$

$NaOH \xrightarrow{NH_4Cl} H_2O + NaCl + NH_3$

$NaOH \xrightarrow{CO_2} Na_2CO_3 + H_2O$

$NaOH \xrightarrow{HCl} NaCl + H_2O$

$NaOH \xrightarrow{NO_2} NaNO_2 + NaNO_3 + H_2O$

$NaOH \xrightarrow{Al_2O_3} NaAlO_2 + H_2O$

$NaOH \xrightarrow{ZnO} Na_2ZnO_2 + H_2O$

$NaOH \xrightarrow{SnO} Na_2SnO_2 + H_2O$

$NaOH \xrightarrow{SnO_2} Na_2SnO_3 + H_2O$

$NaOH \xrightarrow{FeSO_4} Fe(OH)_2(\downarrow) + Na_2SO_4$

$NaOH \xrightarrow{FeCl_3} Fe(OH)_3(\downarrow) + NaCl$

$NaOH \xrightarrow{MnSO_4} \underset{\text{Buff colour}}{Mn(OH)_2(\downarrow)} + Na_2SO_4$

$NaOH \xrightarrow{CuSO_4} \underset{Blue}{Cu(OH)_2} + Na_2SO_4$

$NaOH \xrightarrow{HgCl_2} Hg(OH)_2 \longrightarrow HgO$

$NaOH \xrightarrow{CO} HCOONa$

KOH is mainly used for the absorption of gases like CO_2, SO_2 etc.

KOH + H_2O is called potash lye and is used for making soft soaps

Alcoholic KOH is used as dehydrohalogenating agent in alkyl halides

$C_2H_5Br + Alc.KOH \longrightarrow C_2H_4 + KBr + H_2O$

Oxides

Action in Air

$\text{Alkali Metal} \xrightarrow[O_2]{Air} M_2O \xrightarrow{H_2O} MOH \xrightarrow{CO_2} M_2CO_3$

$Li \xrightarrow[N_2]{O_2} LiO\ \&\ Li_3N$

Oxides: $Na + O_2 \xrightarrow{\Delta} Na_2O\ \&\ Na_2O_2$

$K + O_2 \xrightarrow{\Delta} K_2O, K_2O_2\ \&\ KO_2$

Monoxides contain O^{-2} ion, diamagnetic, colourless and basic

$Na_2O + H_2O \longrightarrow 2NaOH$

$K_2O + H_2O \longrightarrow 2KOH$

$Na_2O + NH_3 \longrightarrow NaNH_2 + NaOH$

Basic nature: CsOH > RbOH > KOH > NaOH > LiOH

Monoxides have antifluorite crystal structures

Peroxides contain $(-O-O-)^{2-}$ ions. Diamagnetic and oxidising agents

$3Na_2O_2 + 2Al \longrightarrow Al_2O_3 + 3Na_2O$

$Na_2O_2 + Cr^{3+} \longrightarrow CrO_4^{2-}$

$Na_2O_2 + CO \longrightarrow Na_2CO_3$

$2Na_2O_2 + 2CO_2 \longrightarrow 2Na_2CO_3 + O_2$

$Na_2O_2 + 2H_2O \longrightarrow 2NaOH + H_2O_2$

$Na_2O_2 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O_2$

used as bleaching agent

Superoxides contain O_2^{-1} ions. Paramagnetic, coloured, oxidising agents.

Stability of superoxides increases with increase in cationic size

$KO_2 > NaO_2$. Larger cation stabilizes larger anion

Contain odd eletron bond [:Ö⋯Ö:]⁻

$KO_2 + 2H_2O \longrightarrow KOH + H_2O_2 + 1/2\ O_2$

$4KO_2 + 2CO_2 \longrightarrow 2K_2CO_3 + 3O_2$

$4KO_2 + 4CO_2 + 2H_2O \longrightarrow 4KHCO_3 + 3O_2$

KO_2 removes CO_2 and produces O_2 which is important in life support systems

Stability order: $K_2O < K_2O_2 < KO_2$

Bond order: $\underset{2}{O_2} > \underset{1.5}{O_2^-} > \underset{1}{O_2^{2-}}$ hence ***B.L*** increases and ***B.E*** decreases

Potassium Permanganate($KMnO_4$)

Preparation:

$\underset{pyrolusite}{2MnO_2} + 4KOH + O_2 \longrightarrow 2K_2MnO_4 + 2H_2O$

$2MNO_2 + 2K_2CO_3 + O_2 \longrightarrow 2K_2MnO_4 + 2CO_2$

$2K_2MnO_4 + Cl_2 \longrightarrow 2KMnO_4 + 2KCl$

$2K_2MnO_4 + H_2O + O_3 \longrightarrow 2KMnO_4 + 2KOH + O_2$

Properties:

It is purple colored crystalline substance

It is well soluble in water

Colour of MnO_4^- is due to charge transfer by oxygen to Mn, because of which Mn changes from +VII to +VI with one unpaired e^- in 3d.

$KMnO_4$ acts as an O.A in acidic, neutral and alkaline media.

Oxidising nature in acidic medium.

$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$

product / reactant		
	X_2 / HX	Fe^{3+} / Fe^{2+}
		NO_3^- / NO_2^-
	I_2 / I^-	S / S^{2-}
	CO_2 / $C_2O_4^{2-}$	SO_4^{2-} / SO_2

denotes reaction with $KMnO_4$

R. K. MALIK'S NEWTON CLASSES, RANCHI

Carbonates and Bicarbonates

Stability Order: $Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3$

$Li_2CO_3 \longrightarrow Li_2O + CO_2$ *Li_2CO_3 is insoluble in water*

$M_2CO_3 + 2H_2O \longrightarrow 2MOH + H_2CO_3$

The solution is alkaline due to Anionic (CO_3^{2-}) hydrolysis

$Na_2CO_3 + CO_2 + H_2O \longrightarrow 2NaHCO_3$

Except Li all others form solid bicarbonates

Sodium carbonate: ($Na_2CO_3.10H_2O$) washing soda

Preparation : LeBlanc Process

$2NaCl + H_2SO_4 \longrightarrow \underset{\text{Salt cake}}{Na_2SO_4} + 2HCl$

$Na_2SO_4 + CaCO_3 + 4C \longrightarrow \underset{\text{Black ash}}{Na_2CO_3 + CaS} + 4CO$

Solvay (Ammonia - Soda) Process

Raw materials are NaCl, NH_3 and $CaCO_3$

$NH_3 + H_2O + CO_2 \longrightarrow NH_4HCO_3$

$NH_4HCO_3 + NaCl \longrightarrow NaHCO_3 + NH_4Cl$

$2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + CO_2 + H_2O$

$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

$CaO + H_2O \longrightarrow Ca(OH)_2$

$2NH_4Cl + Ca(OH)_2 \longrightarrow 2NH_3 + CaCl_2 + 2H_2O$

Sodium Carbonate is white crystalline solid and Efflorescent in air to form $Na_2CO_3 . H_2O$

reactant	product
H_2O	H_2CO_3+NaOH
HCl	NaCl+CO_2+H_2O
SiO_2	Na_2SiO_3+CO_2
SO_2	Na_2SO_3+CO_2
$Ca(OH)_2$	NaOH+$CaCO_3$
$BaCl_2$	$BaCO_3$+NaCl
$(CH_3COO)_2Pb$ / H_2O	$PbCO_3.Pb(OH)_2$+CO_2 +CH_3COONa

denotes reaction with Na_2CO_3

$Na_2CO_3 + K_2CO_3$ is a fusion mixture

$NaHCO_3$ is used as antacid (Eno), in baking powder, in fire extinguishers and in effervescent drinks

K_2CO_3: (Potash or pearl ash)

Prepared by Le Blanc process

Not prepared by Solvay process as $KHCO_3$ is more soluble in water than $NaHCO_3$

K_2CO_3 colourless and deliquescent powder

Used in making soft soap

Sulphates

Na_2SO_4 is called salt cake

$2NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + 2HCl$

$Na_2SO_4.10H_2O$ is called *Glauber's salt*

It is efflorescent in dry air

$Na_2SO_4.10H_2O \longrightarrow Na_2SO_4 + 10H_2O$

Aq.Na_2SO_4 is neutral in nature

$\underset{\text{(Schonite mineral)}}{K_2SO_4.MgSO_4.6H_2O} + 2KCl \longrightarrow 2K_2SO_4 + MgCl_2 + 6H_2O$

$K_2SO_4 + Al_2(SO_4)_3 + H_2O \longrightarrow \underset{\text{potash alum}}{K_2SO_4 . Al_2(SO_4)_3.24H_2O}$

$K_2SO_4 + 4C \xrightarrow{\Delta} K_2S + 4CO$

Except Li_2SO_4 all other sulphates of Na, K, Rb and Cs are soluble in water

Sodium Extraction

Extraction

(1) Castener's process

$\underset{\text{(Fused)}}{NaOH} \rightleftharpoons Na^+ + OH^-$

At Cathode: $Na^+ + e^- \longrightarrow Na$
(Iron)

At Anode: $4OH^- - 4e^- \longrightarrow O_2 + 2H_2O$
(Nickel)

(2) Down's Process

$\underset{\text{(Fused)}}{NaCl} \longrightarrow Na^+ + Cl^-$

At Cathode : $Na^+ + e^- \longrightarrow Na$
(Iron)

At Anode : $2Cl^- - 2e^- \longrightarrow Cl_2$
(Graphite)

$CaCl_2$ and KF are added to the electrolyte to lower the melting point of NaCl

It is a white silvery, soft metal

It gives two bright lines D_1 (589.6 nm) and D_2 (589 nm) in the yellow region of the spectrum

$4C_2H_5Cl + \underset{\text{(Alloy)}}{4Na - Pb} \longrightarrow \underset{\text{TEL}}{Pb(C_2H_5)_4} + 3Pb + 4NaCl$

Action with liquid NH_3: Sodium dissolves in liquid NH_3 to form deep blue coloured solution which is highly conducting due to presence of ammoniated cations and ammoniated electrons.

$Na + (x+y) NH_3 \longrightarrow [Na^+(NH_3)_x] + [e^-(NH_3)_y]$

It is paramagnetic and is a reducing agent.

Alkynes are stereo specifically reduced to Trans- alkenes in presence of Na - liq.NH_3 (Birch reagent)

Permanganate

Oxidising Nature in neutral medium

$MnO_4^- + 2H_2O + 3e^- \longrightarrow MnO_2 + 4OH^-$

$H_2SO_4 + K_2SO_4 + MnO_2 \xleftarrow[H_2O]{MnSO_4} KMnO_4 \xrightarrow[H_2O]{H_2S} S + MnO_2 + KOH + H_2O$

Oxidising nature in alkaline medium.

$MnO_4^- + 2H_2O + 3e^- \longrightarrow MnO_2 + 4OH^-$ dil. Alkaline Medium (Baeyer's Reagent)

$MnO_4^- + e^- \longrightarrow MnO_4^{2-}$ Very Strong Alkaline Medium

$\underset{OH \quad OH}{CH_2-CH_2} \xleftarrow{CH_2=CH_2} KMnO_4 \xrightarrow[H_2O]{KI} KIO_3 + MnO_2 + KOH$

When exposed to sunlight or heated

$2KMnO_4 \longrightarrow K_2MnO_4 + MnO_2 + O_2$

$KMnO_4$ + *conc.* $3H_2SO_4 \longrightarrow K^+ + MnO_3^- + H_3O^+ + 3HSO_4^-$ (*conc.* H_2SO_4 is excess)

$2KMnO_4$ + *conc.* $H_2SO_4 \longrightarrow \underset{\text{explosive oil.}}{Mn_2O_7} + K_2SO_4 + H_2O$ ($KMnO_4$ is excess)

Dichromate

Potassium dichromate ($K_2Cr_2O_7$)

Preparation:

$\underset{\text{(Chromite ore)}}{FeCr_2O_4} + 8Na_2CO_3 + 7CO_2 \xrightarrow{\Delta} 8Na_2CrO_4 + Fe_2O_3 + 8CO_2$

$2Na_2CrO_4 + H_2SO_4 \longrightarrow Na_2Cr_2O_7 + Na_2SO_4 + H_2O$

$Na_2Cr_2O_7 + 2KCl \longrightarrow K_2Cr_2O_7 + 2NaCl$

Properties:

Orange-red coloured crystalline substance.

Moderately soluble in cold water but freely soluble in hot water.

In alkaline solution, Orange colour of $Cr_2O_7^{2-}$ Changes into yellow colour due to formation of CrO_4^{2-}. Again yellow changes to orange in acidic medium.

pH>7: $Cr_2O_7^{2-} + 2OH^- \longrightarrow 2CrO_4^{2-} + H_2O$

pH<7: $2CrO_4^{2-} + 2H^+ \longrightarrow Cr_2O_7^{2-} + H_2O$

$K_2Cr_2O_7 \xrightarrow{HCl} CrCl_3 + Cl_2 + KCl + H_2O$

$K_2Cr_2O_7 \xrightarrow{\Delta} O_2 + Cr_2O_3 + K_2CrO_4$

$K_2Cr_2O_7 \xrightarrow{C_2H_5OH} CH_3CHO \longrightarrow CH_3COOH$

$K_2Cr_2O_7 \xrightarrow{conc.H_2SO_4} KHSO_4 + CrO_3$

$K_2Cr_2O_7 \xrightarrow{H_2SO_4 + NaCl} CrO_2Cl_2$ (chromyl chloride) $\xrightarrow{NaOH} Na_2CrO_4$ (Yellow Solution) $\xrightarrow{(CH_3COO)_2Pb} PbCrO_4$ (yellow)

$K_2Cr_2O_7$ — oxidising nature (in dil H_2SO_4):

$\xrightarrow{I^-} I_2 + Cr^{3+}$

$\xrightarrow{Fe^{2+}} Fe^{3+} + Cr^{3+}$

$\xrightarrow{SO_3^{2-}} SO_4^{2-} + Cr^{3+}$

$\xrightarrow{H_2S} S + Cr^{3+}$

$\xrightarrow{SO_2} SO_4^{2-} + Cr^{3+}$

$\xrightarrow{S_2O_3} SO_4^{2-} + Cr^{3+}$

$\xrightarrow{H_2O_2} CrO_5$ (blue) $\xrightarrow{H_2SO_4} Cr_2(SO_4)_3$

3

R. K. MALIK'S NEWTON CLASSES, RANCHI

Boron & Aluminium

Boron

General Characteristics

General valance Conifiguration is $ns^2\ np^1$

Penultimate shell contains 2,8,18,18,18 e^- respectively

Common oxidation states are +3 and +1

Relative stabilities are

$B^{3+} > Al^{3+} > Ga^{3+} > In^{3+} > Tl^{3+}$
$B^{+} < Al^{+} < Ga^{+} < In^{+} < Tl^{+}$
} This is due to inert pair effect

TlOH forms strongly alkaline solution in water similar to NaOH

Boron has high B.P & M.P due to its covalent polymeric structure.

Boron is a non metal but others are metals.

B exists in crystalline and amorphous forms with icosahedral units with B atoms at all 12 corners. each B is bonded to 5 other B atoms.

B does not form B^{3+} ions aqueous solution due to greater I.E than H.E.

Compounds of B are covalent in nature.

Reactions of B
(at high temperature)

$4B+3O_2 \longrightarrow 2B_2O_3$

$2B+3S \longrightarrow B_2S_3$

$2B+N_2 \longrightarrow 2BN$

$2B+X_2 \longrightarrow 2BX$

$2B+6NaOH \longrightarrow 2Na_3BO_3+3H_2$

$2B+NH_3 \longrightarrow 2BN+3H_2$

$3B+2Mg \longrightarrow Mg_2B_3$

$2B+H_2SO_4 \longrightarrow 2H_3BO_3+3SO_2$

$B+3HNO_3 \longrightarrow H_3BO_3+3NO_2$

No action with HCl

$4B+3CO_2 \longrightarrow 2B_2O_3+3C$

$4B+3SiO_2 \longrightarrow 2B_2O_3+3Si$

Extraction of B
(Amorphous)

a) $Na_2[B_4O_5(OH)_4]8H_2O \xrightarrow{acid} H_3BO_3 \xrightarrow{\Delta} B_2O_3 \xrightarrow{Mg} 2B+3MgO$
Borax

b) $2BCl_3+3H_2 \xrightarrow[\text{W fillament}]{\text{Red hot}} 2B+6HCl$

c) $2BI_3 \xrightarrow[\text{van Arkel}]{\text{Red hot}} 2B+3I_2$

Orthoboric acid

(H_3BO_3) or $B(OH)_3$

Preperation:

a) $Na_2B_4O_7+H_2SO_4+5H_2O \longrightarrow 4H_3BO_3+Na_2SO_4$

b) $Ca_2B_6O_{11}+2SO_2+11H_2O \longrightarrow 6H_3BO_3+2Ca(HSO_3)_2$

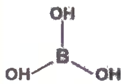
Action of heat:

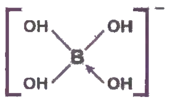
$H_3BO_3 \xrightarrow{100°C} HBO_2 \xrightarrow{160°C} H_2B_4O_7 \xrightarrow{\text{red hot}} B_2O_3$

Orthoboric acid is soluble in H_2O and acts as a monobasic acid. It is a Lewis acid and accepts e^- pair from OH^-

$B(OH)_3+2H_2O \rightleftharpoons H_3O^+ + [B(OH)_4]^-$ $p^{ka} = 9.25$

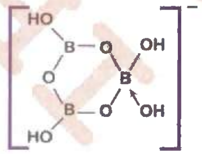
$B(OH)_3+NaOH \rightleftharpoons Na[B(OH)_4] \rightleftharpoons NaBO_2+2H_2O$





At high concentrations polymeric metaborate species are formed.

$3B(OH)_3 \rightleftharpoons H_3O^+ + [B_3O_3(OH)_4]^- + H_2O$ $p^{ka} = 6.85$



H_3BO_3 contains planar trigonal BO_3^{3-} units in which B atoms SP^2 hybridized.

In solid state the $B(OH)_3$ units are hydrogen bonded together in to two dimensional sheets.

Diborane B_2H_6

Preperation:

a) $4BCl_3+3LiAlH_4 \longrightarrow 2B_2H_6+3LiCl+3AlCl_3$

b) $2BCl_3+6H_2 \xrightarrow[\text{discharge}]{\text{Silent electric}} B_2H_6+6HCl$

c) $8BF_3+6LiH \longrightarrow B_2H_6+6LiBF_4$

It is a colourless highly reactive gas

Borax

$Na_2B_4O_7.10H_2O$ or $Na_2[B_4O_5(OH)_4].8H_2O$

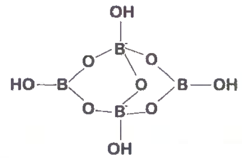
Preparation

a) $4H_3BO_3+Na_2CO_3 \longrightarrow Na_2B_4O_7+6H_2O+CO_2$

b) $Ca_2B_6O_{11}+2Na_2CO_3 \longrightarrow Na_2B_4O_7+2CaCO_3+2NaBO_2$

c) $4NaBO_2+CO_2 \longrightarrow Na_2B_4O_7+Na_2CO_3$

Structure of $[B_4O_5(OH)_4]^{2-}$ anion of Borax



Two Trigonal BO_3 units and two tetrahedral BO_4 units.

Aqueous solution of Borax is alkaline due anionic hydrolysis to Na^+, OH^- and H_3BO_3.

Hence Borax can be titrated as an alkali against a strong acid using methyl orange indicator.

$Na_2[B_4O_5(OH)_4].8H_2O+2HCl \longrightarrow 2NaCl+4H_3BO_3+5H_2O$

or

$[B_4O_5(OH)_4]^{2-}+5H_2O \rightleftharpoons 2B(OH)_3+2[B(OH)_4]^-$

$2[B(OH)_4]^-+2H_3O^+ \longrightarrow 2B(OH)_3+4H_2O$

Borax is also used as a buffer since its aqueous solution contains equal amounts of weakacid and its salt.

Borax swells on strong heating and then sets to a glassy bead on cooling.Fusion with metallic salts (which forms oxides on heating) forms characteristic coloured metaborates.

$Na_2B_4O_7.10H_2O \xrightarrow{\Delta} 2NaBO_2+B_2O_3+10H_2O$

$CuO+B_2O_3 \longrightarrow Cu(BO_2)_2$ blue bead

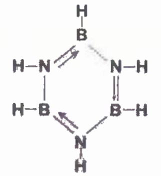
This reaction provides a means of identifying the transition metal

Boron

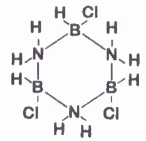
Reactions of B_2H_6

- $B_2H_6 \xrightarrow{\text{red hot}} B + H_2$
- $B_2H_6 \xrightarrow{O_2,\Delta} B_2O_3 + H_2O$
- $B_2H_6 \xrightarrow{H_2O} H_3BO_3 + H_2$
- $B_2H_6 \xrightarrow{HCl} B_2H_5Cl + H_2$
- $B_2H_6 \xrightarrow{Cl_2} BCl_3 + HCl$
- $B_2H_6 \xrightarrow{LiH} 2LiBH_4$
- $B_2H_6 \xrightarrow{CO} BH_3CO$
- $B_2H_6 \xrightarrow{Me_3N} Me_3NBH_3$
- $B_2H_6 \xrightarrow{NH_3} B_2H_6.2NH_3 \xrightarrow{\Delta} B_3N_3H_6 + H_2$

In-organic benzene

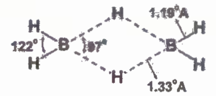


$B_3N_3H_6 + 3HCl \longrightarrow B_3N_3H_9Cl_3$



Both N and B atoms are SP^2 hybridized

Structure



Each B-H-B bond is 3 centerd $2e^-$ bond

B atom is sp^3 hybridized.

Each B-H-B is sp^3-s-sp^3 overlap of orbitals

Boron Halides BX_3

Preparation:

a) $B_2O_3 + 3C + 3X_2 \longrightarrow 2BX_3 + 3CO$

b) $2B + 3X_2 \xrightarrow{\Delta} 2BX_3$

Covalent in nature, electron deficient Lewis acidic nature

$BF_3 < BCl_3 < BBr_3 < BI_3$

Except BF_3, the others are hydrolysed in water.

$BCl_3 + 3H_2O \longrightarrow H_3BO_3 + 3HCl$

Boron Nitride

Preparation

a) $2B + N_2 \longrightarrow 2BN$

b) $2B + 2NH_3 \longrightarrow 2BN + 3H_2$

c) $Na_2B_4O_7 + 2NH_4Cl \longrightarrow 2BN + 2NaCl + B_2O_3 + 4H_2O$

d) $BCl_3 + 6NH_3 \longrightarrow B(NH)_3 + 3NH_4Cl$

$2B(NH_2)_3 \xrightarrow{\Delta} B_2(NH)_3 + 3NH_3$

$B_2(NH)_3 \longrightarrow 2BN + NH_3$

It is a white powder, M.P = 3000°C

Chemically inert

Do not react with mineral acids

Dissolves in HF $BN + 4HF \longrightarrow NH_4BF_4$

No action with alkalies, but when fuses with K_2CO_3 forms KCNO and KBO_2

$BN + K_2CO_3 \longrightarrow KBO_2 + KCNO$

It's structure is similar to graphite

Boron trioxide (B_2O_3)

Preperation:

$4B + 3O_2 \xrightarrow{\Delta} 2B_2O_3$

$2H_3BO_3 \xrightarrow{\Delta} B_2O_3 + 3H_2O$

It is anhydride of orthoboric acid.

It is an acidic oxide

Reduced by Mg to B on heating

$B_2O_3 + 3Mg \longrightarrow 2B + 3MgO$

It behaves as a base in presence of stronger acidic oxides

$2B_2O_3 + P_4O_{10} \longrightarrow 4BPO_4$

Aluminium

Alums

M.F = $M_2SO_4.T_2(SO_4)_3.24H_2O$; E.F = $MT(SO_4)_2.12H_2O$

M = Na^+, K^+, Rb^+, Cs^+, NH_4^+, Tl^+ : T = Al^{3+}, Cr^{3+}, Fe^{3+}, Mn^{3+}, Co^{3+}

Acts as coagulants, germicides, antiseptics and mordants

$AlCl_3$

White powder. Covalent when anhydrous

It is deliquescent and fumes in moist air due to hydrolysis $AlCl_3 + 3H_2O \longrightarrow Al(OH)_3 + 3HCl$

It exists as dimer in solid state. (Al_2Cl_6) and in non polar solvents

The dimer is broken in aq.solution as $[Al(H_2O)_6]^{3+}$ and $3Cl^-$ ions due to greater hydration energy

It's aq.solution is acidic due to cationic hydrolysis

$AlCl_3$ is electron deficient and is a Lewis acid

It is used as a catalyst in Friedel - crafts reactions

Al_2O_3

White solid, not soluble in water

$Al_2(SO_4)_3 \xrightarrow{\Delta} Al_2O_3 + 3SO_3$

$2AlF_3 + B_2O_3 \longrightarrow Al_2O_3 + 2BF_3$

Amphoteric Oxide: $\xrightarrow{HCl} AlCl_3 + H_2O$; $\xrightarrow{NaOH} NaAlO_2 + H_2O$

Mixed with metallic oxides it forms coloured stones used in jewellery.

eg. Ruby(red), sapphire(blue), Emery(green), Amethyst(violet), used as refractory material, abrasive(Alundum)

Extraction

Extraction is done from Bauxite

$Al_2O_3.2H_2O$ (impurities Fe_2O_3 → Red ; SiO_2 → White)

- Red: Baeyers: NaOH → $NaAlO_2 \xrightarrow{H_2O} Al(OH)_3$
- Red: Hall's: Na_2CO_3 → $NAlO_2$ → $Al(OH)_3$
- White: Serpeck's: C, N_2 → AlN + Si + CO $\xrightarrow{H_2O}$ $Al(OH)_3$

$Al(OH)_3 \xrightarrow{\text{Ignition}} Al_2O_3$(Alumina)

Hall Heroult Electrolytic reduction

$Al_2O_3 + Na_3AlF_6 + CaF_2$ = Electrolyte

$Na_3AlF_6 \rightleftharpoons AlF_3 + 3NaF$

$AlF_3 \longrightarrow Al^{+3} + 3F^-$

At Cathode : $Al^{+3} + 3e^- \longrightarrow Al$

At Anode : $2F^- - 2e^- \longrightarrow F_2$

$2Al_2O_3 + 6F_2 \longrightarrow 4AlF_3 + 3O_2$

Purification of Al done by Hoope's process

Reactions

- $Al \xrightarrow{N_2} AlN \xrightarrow{H_2O} Al(OH)_3$
- $Al \xrightarrow{Fe_2O_3} Al_2O_3 + Fe$
- $Al \xrightarrow{X_2} AlX_3 \xrightarrow{H_2O} Al(OH)_3 + HX$
- $Al \xrightarrow{O_2} Al_2O_3$
- $Al \xrightarrow{H_2O} Al(OH)_3 + H_2$
- $Al \xrightarrow{HCl} AlCl_3 + H_2$
- $Al \xrightarrow{Conc.H_2SO_4} Al_2(SO_4)_3 + SO_2 + H_2O$
- $Al \xrightarrow{NaOH} NaAlO_2 + H_2$

Alloys of Al:

Magnalium: Al + Mg

Aluminium Bronze: Cu + Al + Sn

Y Alloy: Al + Cu + Ni + Mg

Duralumin: Al + Cu + Mg + Mn

Nickeloy: Al + Cu + Ni

Alnico: Steel + Al + Ni + Co

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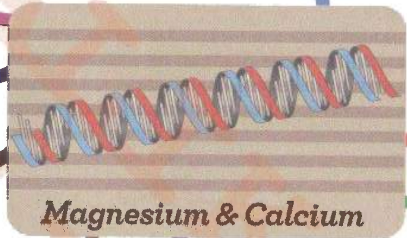


General Characteristics

- Be,Mg,Ca,Sr,Ba and Ra - Alkaline Earth Metals
- Valence Configuration : ns^2 & +2 Oxidation state
- Harder than alkali metals and hardness decreases down the group
- Atomic and ionic radii increases from Be to Ra but smaller than I A
- Denser than IA.Decreases upto Ca and then increases,
  Ra > Ba > Sr > Be > Mg > Ca
  5.5 3.62 2.63 1.84 1.74 1.55 (g cm^{-1})
- Less electroposite than I A. Electropositive nature increases from Be to Ra
- Heats of hydration of M^{2+} ions decreases from Be^{2+} to Ba^{2+} due to increase in size of M^{2+} ion
  (Smaller the size greater the hydration energy)
- H.E of *IIA*, M^{2+} ions > H.E of *IA*, M^+ ions.
- Ionic mobility and hence ionic conductance increases from $[Be(aq)]^{2+}$ to $[Ba(aq)]^{2+}$ due to decrease in ionic radii
- Be and Mg exhibit no flame colouration but others Ca(brick red), Ba(apple green),Sr & Ra(crimson red)
- Reducing nature of II A metals < I A metals
- Reducing agent nature Be < Mg < Ca < Sr < Ba

Chemical Properties

Action with water

$M + 2H_2O \longrightarrow M(OH)_2 + H_2$

Ca, Sr, Ba decompose cold water but Mg decomposes boiling water and Be not even steam

Oxides and Hydroxides

$2M + O_2 \xrightarrow{\Delta} 2MO$ - Monoxide

Ba and Ra even form peroxides

$2Ba + O_2 \xrightarrow{\Delta} 2BaO \xrightarrow[\Delta]{O_2} 2BaO_2$ - Peroxide

- BeO is covalent and amphoteric but others are ionic and basic, which increases down the group
- BeO and MgO are refractory materials
- $MgO + H_2O \longrightarrow Mg(OH)_2$ + Heat

- Basic nature of hydroxides increases in the order
  $Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$
- Thermal stability and solubility in water increases in the order
  $Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$
- $\underset{\text{(Lime water)}}{Ca(OH)_2} + CO_2 \longrightarrow \underset{\text{milky}}{CaCO_3\downarrow} + H_2O \xrightarrow{CO_2} \underset{\text{Soluble}}{Ca(HCO_3)_2}$
- $\underset{\text{(Baryta water)}}{Ba(OH)_2} + CO_2 \longrightarrow \underset{\text{milky}}{BaCO_3\downarrow} + H_2O$
- $Mg(OH)_2$ readily dissolves in NH_4Cl solution
  $Mg(OH)_2 + 2NH_4Cl \rightleftharpoons MgCl_2 + 2NH_4OH$
- $Mg(OH)_2$ in H_2O is a suspension under the name Milk of Magnesia an antacid
  Slaking of lime : $\underset{\text{Quick lime}}{CaO} + H_2O \longrightarrow \underset{\text{Slaked lime}}{Ca(OH)_2}$ + Heat

Carbonates and Bicarbonates

- Thermal stability and ionic nature increases with the increases in cationic size
- $BeCO_3 < MgCO_3 < CaCO_3 < SrCO_3 < BaCO_3$
  (Larger cation stabilises larger anion)
- $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$
- Solubility in water decreases in the order
  $BeCO_3 > MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$
- More soluble in solution of CO_2 than in H_2O
  $\underset{\text{(insoluble)}}{CaCO_3} + H_2O + CO_2 \longrightarrow \underset{\text{(soluble)}}{Ca(HCO_3)_2}$
- Bicarbonates are known in solution only and decomposes on heating
  $Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3\downarrow + CO_2 + H_2O$
- $Ca(HCO_3)_2$ and $Mg(HCO_3)_2$ exist in water causes temporary hardness, which can be removed by boiling
- $Mg(HCO_3)_2$ and $Ca(HCO_3)_2$ are amphoteric
  $Ca(HCO_3)_2 + H_2SO_4 \longrightarrow CaSO_4 + 2H_2O + 2CO_2$
  $Ca(HCO_3)_2 + Ca(OH)_2 \longrightarrow 2CaCO_3\downarrow + 2H_2O$

Sulphates

- $MgO + H_2SO_4 \longrightarrow MgSO_4 + H_2O$
- $Mg(OH)_2 + H_2SO_4 \longrightarrow MgSO_4 + 2H_2O$
- $MgCO_3 + H_2SO_4 \longrightarrow MgSO_4 + H_2O + CO_2$
- $CaCl_2 + Na_2SO_4 \longrightarrow CaSO_4 + 2NaCl$
- Thermal stability order :
  $BeSO_4 < MgSO_4 < CaSO_4 < SrSO_4 < BaSO_4$
  $MSO_4 \xrightarrow{\Delta} MO + SO_3$ (or) $MO + SO_2 + O_2$
- Solubility of sulphates in water decreases in the order
  $BeSO_4 > MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$
- $MSO_4 + 4C \xrightarrow{\Delta} MS + 4CO$
- $MgSO_4 \cdot 7H_2O$ is Epsom salt, Efflorescent and purgative
- $\underset{\substack{\text{Gypsum}\\\text{(Monoclinic)}}}{CaSO_4 \cdot 2H_2O} \xrightarrow{\Delta} \underset{\text{orthorhombic}}{CaSO_4 \cdot 2H_2O} \xrightarrow{120^\circ C} \underset{\text{pop}}{CaSO_4 \cdot 1/2\, H_2O}$
  $\xrightarrow{200^\circ C} \underset{\text{anhydrite}}{CaSO_4} \xrightarrow[\text{heated}]{\text{Strongly}} \underset{\text{(lime)}}{CaO} + SO_2 + O_2$
- Setting of pop takes place in two stages
  $\underset{\text{Pop}}{2\,CaSO_4 \cdot H_2O} \xrightarrow[H_2O]{\text{Setting}} \underset{\substack{\text{Orthorhombic}\\\text{gypsum}}}{CaSO_4 \cdot 2H_2O} \xrightarrow{\text{Hardening}} \underset{\substack{\text{Monoclinic}\\\text{gypsum}}}{CaSO_4 \cdot 2H_2O}$
- $BaSO_4$ + ZnS is Lithophone, used as white paint

Halides

- $Mg + 2HCl \longrightarrow MgCl_2 + H_2$
- $CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$
- $BeCl_2$ is covalent polymer and other chlorides are ionic
- Halides are hygroscopic and form hydrates like $MgCl_2 \cdot 6H_2O$, $CaCl_2 \cdot 6H_2O$ etc.
- BeF_2 is soluble in water but other fluorides are insoluble
- Chlorides,Bromides and Iodides of Mg,Ca,Sr,Ba are readily soluble in water
- Solubility of chlorides in H_2O decreases in the order $MgCl_2 > CaCl_2 > SrCl_2 > BaCl_2$
- $CaCl_2$ is a dehydrating agent and is used for treating ice on roads

Extraction of Magnesium

From Magnesite

$MgCO_3 \xrightarrow{\Delta} MgO + CO_2$

$MgO + C + Cl_2 \longrightarrow MgCl_2 + CO$

From Carnalite:

$KCl \cdot MgCl_2 \cdot 6H_2O \xrightarrow[\text{gas}]{HCl} MgCl_2 + KCl + 6H_2O$

Electrolysis

Fused $MgCl_2 \rightleftharpoons Mg^{2+} + 2Cl^-$

($NaCl + CaCl_2$)

At Cathode(steel) : $Mg^{2+} + 2e^- \longrightarrow Mg$

At Anode (graphite) : $2Cl^- - 2e^- \longrightarrow Cl_2$

Coal gas is blown to prevent the oxidation of Mg produced

Properties of Mg

- Mg is silvery white metal. M.P is 650°C
- Good conductor of heat and electricity

Mg +	Product
Air O_2 & N_2	$MgO + Mg_3N_2$
H_2O	$Mg(OH)_2 + H_2$
dil.H_2SO_4	$MgSO_4 + H_2$
Conc.H_2SO_4	$MgSO_4 + SO_2 + H_2O$
R - X	R - MgX
CO_2	MgO + C
SO_2	MgO + S
NO	$MgO + Mg_3N_2$
B_2O_3	MgO + B
$FeCl_3$ Solution	$MgCl_2 + Fe(OH)_3 + H_2$
NH_4Cl Solution	$MgCl_2 + NH_3 + H_2$

Alloys of Mg

Magnalium :
Al (95%) + Mg (5%)

Elektron :
Mg (95%) + Zn (4.5%) + Cu (0.5%)

Duralumin :
Al (95%) + Mg (0.5%) + Mn (0.5%) & Cu (4%)

General Characteristics

- C, Si, Ge, Sn and Pb
- General Valance configuration is $ns^2 np^2$
- Common O.S are +4 and +2 (Inert pair effect)
- C and Si exhibits +4 stable state
- $Ge^{4+} > Sn^{4+} > Pb^{4+}$ but $Ge^{2+} < Sn^{2+} < Pb^{2+}$
- M^{2+} compounds are Ionic and reducing agents
- M^{4+} compounds are Covalent and oxidising agents
- Maximum covalency of C is 4 but others can exibit 6 by utilising vacant d-orbitals

  $SiF_4 + 2F^- \rightarrow [SiF_6]^{2-}$
- Except Pb all the others show Allotropy
- Bond Energy and catenation property order

  C - C > Si - Si > Ge - Ge > Sn - Sn
- Stability of Hydrides

  $CH_4 > SiH_4 > GeH_4 > SnH_4 > PbH_4$
- Stability of Halides

  $CCl_4 > SiCl_4 > GeCl_4 > SnCl_4 > PbCl_4$

  $CF_4 > CCl_4 > CBr_4 > CI_4$
- Acidic nature of dioxides

  $CO_2 > SiO_2 > GeO_2 > SnO_2 > PbO_2$

Silicones

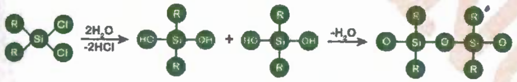
Silicones : These are organosilicon polymers contaning -Si-O-Si- linkages and R_2SiO group as the unit building block (R = Methyl or Phenyl)

Manufacture:

a) Formation of alkyl cholrosilanes

$SiCl_4 + RMgCl \rightarrow RSiCl_3 \rightarrow R_2SiCl_2 \rightarrow R_3SiCl \rightarrow R_4Si$

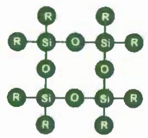
b) Hydrolysis of alkyl chlorosilanes followed by condensation polymerisation.



Poly dimethyl Siloxane (Silicone oil)

- Cyclic Structures, with rings containing 3,4,5 or 6 Si atoms are obtained on carefull controlled hydrolysis.

tricyclo dimethyl siloxane



tetrkis cyclo-dimethylsiloxane

- Hydrolysis of $RSiCl_3$ gives a cross linked polymer.

Because of high B.E of Si-O & Si-C bonds, silicones are highly inert, strongly water repellent, insulators and have non stick and anti-foaming properties.

Oxides of C & Si

CO : Neutral oxide. Colourless and Odourless , poisonous gas

- Preparartion: a) $HCOOH \xrightarrow[-H_2O]{conc.\ H_2SO_4} CO$

  b) $H_2C_2O_4 \xrightarrow[-H_2O]{conc.\ H_2SO_4} CO + CO_2$

  b) $ZnO + C \xrightarrow{\Delta} Zn + CO$
- Structure: $:C \equiv O:$ C-atom is SP hybridized. B O = 3. C-O bond length = 1.13 A°
- The lone pair on C-atom is donated to certain metals to form $M \leftarrow CO$ carbonyls.

viz.	$Ni(CO)_4$	$Fe(CO)_5$	$Cr(CO)_6$	$V(CO)_6$
	sp^3	dsp^3	d^2sp^3	d^2sp^3
	Dia	Dia	Dia	para
	T.H	T.B.P	octahedral	octahedral

- O.S of the metal is zero & EAN is 36 (kr)
- CO forms carboxy heamoglobin with blood which leads to fatal.
- It is a good reducing agent

  $ZnO + CO \rightarrow Zn + CO_2$ | $PbO + CO \rightarrow Pb + CO_2$

  $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$ | $I_2O_5 + CO \rightarrow I_2 + 5CO_2$

CO_2 : It is an acidic oxide. Colourless & Odourless gas.

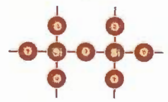
- Preparartion: a) $Na_2CO_3 + 2HCl \longrightarrow 2NaCl + H_2O + CO_2$

  b) $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

  c) $2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + H_2O + CO_2$
- It turns lime water into milky

  $Ca(OH)_2 + CO_2 \rightarrow \underset{milky}{CaCO_3} + H_2O \xrightarrow{CO_2} \underset{soluble}{Ca(HCO_3)_2}$
- $CO_2 + H_2O \rightleftharpoons H_2CO_3$ unstable acid
- H_2CO_3 is a diabasic acid → $NaHCO_3$ acidic salt; Na_2CO_3 normal salt
- Structure: $O=C=O \leftrightarrow \overset{+}{O} \equiv C-\overset{-}{O} \leftrightarrow \overset{-}{O}-C \equiv \overset{+}{O}$
- C-atom is sp hybridized, bond order=2, μ = 0, C-O bond length = 1.15A°

Structure of SiO_2



SiO_2 (Silica) It is an acidic oxide

- It is widely found as Sand and Quartz
- The main forms of SiO_2 are quartz , tridymite & crystobalite
- Coloured quartz are used as gems Amethyst(violet), rose quartz(pink) etc
- Flint, opal, agate, onyx & jasper are amorphous silica
- Kieselguhr is siliceous rock composed of remains of minute sea organisms.
- SiO_2 is soluble in HF

  $SiO_2 + 4HF \rightarrow SiF_4 \xrightarrow{2HF} H_2SiF_6$
- $SiO_2 + 2NaOH \rightarrow \underset{water\ glass}{Na_2SiO_3} + H_2O$
- $Na_2SiO_3 + 2HCl \rightarrow \underset{(silicic\ acid)}{H_2SiO_3} + 2NaCl$
- SiO_2 is a 3D polymer in which each Si atom (SP^3) is bonded with 4 O-atoms tetrahedraly.
- Silica gel is a granular, vitreous, highly porous form of silica made synthetically from sodium silicate & is used as drying agent.

SiC Carborundum or Artificial Diamond

$SiO_2 + 3C \xrightarrow{\Delta} SiC + 2CO$

- Very hard (Diamond > B_4C > SiC). Chemically inert.
- Soluble in alkali in presence of oxygen

  $SiC + O_2 + 4NaOH \rightarrow Na_2SiO_3 + Na_2CO_3 + H_2O$
- Used as abrasive.

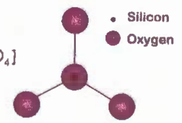
Silicates

- Silicate minerals together with silica and aluminosilicate clay make 95% of the Earth's crust.
- They contain different modes of combination of $(SiO_4)^{4-}$ tetrahedral units.
- The Si-O bond is 50% ionic & 50% covalent (E.N difference = 3.5 - 1.8 = 1.7).
- The radius of Si^{4+} (0.4 A°) & O^{2-}(1.4 A°) makes the radius-ratio 0.29 with co-ordination number 4 around Si.
- The $(SiO_4)^{4-}$ tetrahedral units may be single or share 1, 2, 3 or 4 O-atoms through corners only giving rise to various types.

Types of Silicates:

1. **Orthosilicates:** (Neso-silicates). No O-atom is shared. Contain discrete SiO_4^{4-} units with general formula $M_2^{II}SiO_4$.

   eg: Zircon = $ZrSiO_4$
   Olivine = $(Mg.Fe)_2.[SiO_4]$
   Fenacite = Be_2SiO_4
   Forsterite = Mg_2SiO_4
   Williemite = Zn_2SiO_4

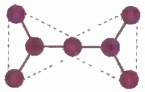


2. **Pyrosilicates (Soro-silicates)**

   One O-atom is shared at one corner. Contain discrete $(Si_2O_7)^{6-}$ unit.

   eg: Thortveitite: $Sc_2Si_2O_7$
   Hemimorphite: $Zn_4Si_2O_7(OH)_2.H_2O$

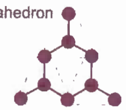
   Pyrosilicates are rare.



3. **Ring (cyclic) Silicates :**

   Two O-atoms are shared per tetrahedron contain $(SiO_3)_n^{2n-}$ units

   eg: Benitoite : $BaTiSi_3O_9$
   Beryl : $Be_3Al_2(SiO_3)_6$

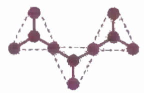


4. **Chain Silicates:**

   a) single chain (pyroxenes)

   Two O-atoms are shared per tetrahedron contain $(SiO_3)_n^{2-}$ units

   eg: Spodumene : $LiAl(SiO_3)_2$
   Diopside : $CaMg(SiO_3)_2$
   Enstatite : $Mg_2(SiO_3)_2$
   Wollastonite : $Ca_3(SiO_3)_3$



   b) Double Chains (amphiboles)

   contain $(Si_4O_{11})_n^{6n-}$ units. Asbestos minerals. Some tetrahedron share two and some share three corners ($2\frac{1}{2}$ 0 - atoms)

   eg: Tremolite : $Ca_2Mg_5(Si_4O_{11})_2(OH)_2$
   Crocidolite : $Na_2Fe_3^{II}Fe_2^{III}(Si_4O_{11})_2(OH)_2$

5. **Sheet Silicates (phyllo - silicates)**

   Three O-atoms are shared at three corners of the tetrahedron. Contains $(Si_2O_5)_n^{2n-}$

   eg: Muscovite : $KAl_2(OH)_2AlSi_3O_{10}$ Mica
   Talc : $Mg_3(OH)_2Si_4O_{10}$
   Kaolin: $Al_2(OH)_4Si_2O_5$ Clay

6. **Three Dimensional Silicates**

   Four O-atoms are shared from all corners of SiO_4 tetrahedron resulting in 3D lattice of formula SiO_2

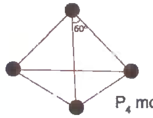
   eg: Quartz : $(SiO_2)_n$

   Due to replacement of Si atoms form
   a) Orthoclase feldspar $KAlSi_3O_8$
   b) Zeolite : $Na_2Al_2Si_3O_8.xH_2O$
   c) Ultramarine : $Na_8[(AlSiO_4)_6]S_2$

R. K. MALIK'S NEWTON CLASSES, RANCHI

General characterstics

1. N, P, As, Sb and Bi - p block.
2. ns^2np^3, common oxidation states are -3,+3,+5.
3. O.S of N are -3(NH_3), -2(NH_2NH_2), -1(NH_2OH), 0(N_2),+1(N_2O),+2(NO),+3(N_2O_3),+4(NO_2),+5(N_2O_5, HNO_3)
4. N,P (non metals) , As (Metalloid), and Sb, Bi (Metal).
5. Atomic radii and Density order
   N < P < As < Sb < Bi
6. I.E and E.N order
   N > P > As > Sb > Bi
7. N_2(N≡N) diatomic , while P, As, Sb are tetra atomic (P_4, As_4, Sb_4) and tetrahedral with a bond angle of 60°
8. Maximum covalency of N is 4.
9. Basic Nature of hydrides
   $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$



P_4 molecule

1. N_2O (Laughing gas)

$NH_4NO_3 \longrightarrow N_2O + 2H_2O$

It is decomposed by red hot copper.

Structure

$N\equiv\overset{+}{N}\rightarrow O^- \longleftrightarrow \overset{-}{N}=\overset{+}{N}=O$

2. NO (Nitric Oxide)

$3Cu + 8HNO_3 \text{(dil)} \longrightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$

$N_2 + O_2 \xrightarrow{\text{Electric Arc}} 2NO$

$4NH_3 + 5O_2 \xrightarrow[\Delta]{Pt} 4NO + 6H_2O$

It dissolves in $FeSO_4$ solution to form a nitrosyl complex (brown ring test)

It is both oxidising as well as reducing agent.

Structure:

$:\dot{N}=O:$ *paramagnetic*

3. N_2O_3 (Dinitrogen trioxide):

Acidic oxide . Anhydride of HNO_2

$N_2O_3 + H_2O \longrightarrow 2HNO_2$

$N_2O_3 + NaOH \longrightarrow 2NaNO_2 + H_2O$

It is an unstable oxide.

Structure: O = N – O – N = O

4. NO_2 (Nitrogen dioxide)

$Pb(NO_3) \xrightarrow{\Delta} 2PbO + 4NO_2 + O_2$

$Cu + 4HNO_3 \text{(conc)} \longrightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$

It is mixed anhydride

$2NO_2 + H_2O \longrightarrow HNO_2 + HNO_3$

It is both oxidising agent and reducing agent.

it dimerises to N_2O_4 solid

$2NO_2 \rightleftharpoons N_2O_4$

(brown gas) (paramagnetic) — (colourless solid) (diamagnetic)

5. N_2O_5 (Di Nitrogen pentoxide)

$2HNO_3 + P_2O_5 \longrightarrow N_2O_5 + 2HPO_3$

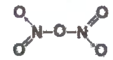
It is anhydride of HNO_3

$N_2O_5 + H_2O \longrightarrow 2HNO_3$

It is strong oxidising agent

$N_2O_5 + NaCl \longrightarrow \overset{+}{Na}\overset{-}{NO_3} + \overset{+}{NO_2}\overset{-}{Cl}$

Structure:

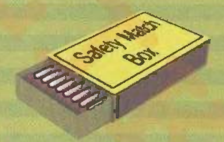


N_2O, NO, N_2O_3, NO_2, and N_2O_5

neutral — Increasing Acidic Strength

Oxides of Nitrogen

General Characteristics



Nitrogen & Phosphorus

Oxyacids of Nitrogen

HNO_2: Nitrous acid

$NaNO_2 + H_2SO_4 \longrightarrow Na_2SO_4 + 2HNO_2$

$NaNO_3 + Pb \xrightarrow{\Delta} NaNO_2 + PbO$

$N_2O_3 + H_2O \longrightarrow 2HNO_2$

weak acid. Forms nitrites with alkalies

$HNO_2 + NaOH \longrightarrow NaNO_2 + H_2O$

unstable, undergoes auto-oxidation

$3HNO_2 \longrightarrow HNO_3 + 2NO + H_2O$

oxidising in nature

$2HNO_2 \longrightarrow 2NO + H_2O + (O)$

reducing in nature

$HNO_2 + (O) \longrightarrow HNO_3$

$HNO_2 + R{-}NH_2 \longrightarrow R{-}OH + N_2 + H_2O$

$2HNO_2 + NH_2CONH_2 \longrightarrow 2N_2 + CO_2 + 3H_2O$

$C_6H_5NH_2 + HNO_2 \xrightarrow[0^\circ]{HCl} C_6H_5N_2Cl + 2H_2O$

Structure:

H–O–N=O & H–N(=O)=O *Tautomers*

H^+/I^-	Fe^{2+}	H_2S
I_2	Fe^{3+}	S
SO_2	reactant	Br_2
SO_4^{2-}	product	HBr
MnO_4^-/H^+	$Cr_2O_7^{2-}/H^+$	H_2O_2
Mn^{2+}	Cr^{3+}	H_2O

denotes reaction with HNO_2

Oxidising Agent

Reducing Agent

Nitric acid:

HNO_3: (Aqua Fortis)

$KNO_3 + H_2SO_4 \longrightarrow HNO_3 + KHSO_4$

Birkeland Eyde process:

$N_2 + O_2 \longrightarrow 2NO$ $\Delta H = +43.20$ kcal

$2NO + O_2 \longrightarrow 2NO_2$

$2NO_2 + H_2O \longrightarrow HNO_2 + HNO_3$

$3HNO_2 \longrightarrow HNO_3 + H_2O + 2NO$

Ostwald's Process:

$4NH_3 + 5O_2 \xrightarrow{Pt} 4NO + 6H_2O$ $\Delta H = -21.5$ k.cal

$2NO + O_2 \longrightarrow 2NO_2$

$4NO_2 + 2H_2O + O_2 \longrightarrow 4HNO_3$

Properties of HNO_3:

HNO_3 is a powerful oxidising agent; it oxidises almost all metals ,non-metals and many compounds.

It gets reduced as:

$\underset{+5}{HNO_3} \longrightarrow \underset{+4}{NO_2} \longrightarrow \underset{+2}{NO} \longrightarrow$

$\underset{+1}{N_2O} \longrightarrow \underset{0}{N_2} \longrightarrow \underset{-3}{NH_3}$

greater the change in Oxidation state, weaker the oxidising action of HNO_3

Non Metals are oxidised to their higher oxyacids and HNO_3 is reduced to NO_2

C ⟶ H_2CO_3

S ⟶ H_2SO_4

P ⟶ H_3PO_4

I_2 ⟶ HIO_3

Oxidation of compounds

$SO_2 \longrightarrow H_2SO_4$

$H_2S \longrightarrow S$

$HBr \longrightarrow Br_2$

$KI \longrightarrow I_2$

oxidation of metal depends mainly on

a) nature of metal

b) conc. of acid

noble metals (Au, Pt) dissolve in Aqua regia (HNO_3 +3HCl)

$HNO_3 + 3HCl \longrightarrow NOCl + 2H_2O + 2Cl$

$Au + 3Cl \longrightarrow AuCl_3 \xrightarrow{HCl} HAuCl_4$

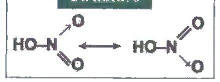
Nitration Mixture:

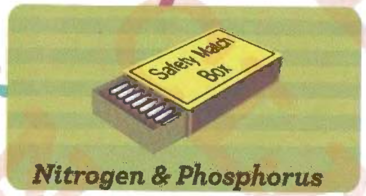
$HNO_3 + 2H_2SO_4 \xrightarrow{\text{nitration}} \overset{+}{NO_2} + 2H\overset{-}{SO_4} + \overset{+}{H_3O}$

Action with Metals:

HNO_3	Metal	Main Product
Very Dilute	Mg, Mn	$H_2 + MNO_3$
	Fe, Zn, Sn	$NH_4NO_3 + MNO_3$
Dilute	Pb, Cu, Ag, Hg	$NO + MNO_3$
	Fe , Zn	$N_2O + MNO_3$
	Sn	$NH_4NO_3 + Sn(NO_3)_2$
Concentrated	Zn, Pb, Cu, Ag	$NO_2 + MNO_3$
	Sn	$NO_2 + H_2SnO_3$
	Fe, Ni, Co, Cr, Al	Passive (inert)

Structure





Phosphorous — Oxy-Acids, Phosphine, Oxides, Ammonia

Oxyacids of Phosphorus:

1. Phosphorus acids: (H_3PO_3)

$P_4O_6 + 6H_2O \longrightarrow 4H_3PO_3$

$4H_3PO_3 \xrightarrow{200^\circ C} 3H_3PO_4 + PH_3$

Dibasic acid and forms NaH_2PO_3 (acidic salt) and Na_2HPO_3 (normal salt)

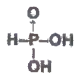
It is a reducing agent

$H_3PO_3 + H_2O \longrightarrow H_3PO_4 + 2[H]$

Reduces

$CuSO_4 \longrightarrow Cu$	$I_2 \longrightarrow HI$
$AgNO_3 \longrightarrow Ag$	$KMnO_4 / H^+ \longrightarrow MnSO_4$
$HgCl_2 \longrightarrow Hg_2Cl_2$	

Structure:



2. Orthophosporic Acid (H_3PO_4):

$P_4O_{10} + 6H_2O \longrightarrow 4H_3PO_4$

$PCl_5 + 4H_2O \longrightarrow H_3PO_4 + 5HCl$

$P + 5HNO_3 \longrightarrow H_3PO_4 + 5NO_2 + H_2O$

$Ca(PO_4)_2 + 3H_2SO_4 \longrightarrow 2H_3PO_4 + 3CaSO_4$

Tribasic Acid and forms 3 series of salts

NaH_2PO_4, Na_2HPO_4, Na_3PO_4
(Acidic) (Acidic) (Normal salt)

$2H_3PO_4 \xrightarrow{250^\circ C} H_4P_2O_7 + H_2O$

$\quad \xrightarrow{\Delta} 2HPO_3 + H_2O$

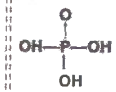
$H_3PO_4 + 3AgNO_3 \longrightarrow Ag_3PO_4 + 3HNO_3$ *(yellow ppt)*

$H_3PO_4 + 3BaCl_2 \longrightarrow Ba_3(PO_4)_2 + 6HCl$ *(white ppt)*

$H_3PO_4 + 3NaBr \longrightarrow Na_3PO_4 + 3HBr$

$MgSO_4 + NH_4Cl + H_3PO_4 \longrightarrow Mg(NH_4)PO_4\downarrow + H_2SO_4 + HCl$ *(white ppt test for Mg^{+2})*

Structure:



Phosphine (PH_3)

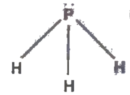
$P_4 + 3NaOH + 3H_2O \longrightarrow 3NaH_2PO_2 + PH_3$

$4H_3PO_3 \xrightarrow{\Delta} 3H_3PO_4 + PH_3$

$Ca_3P_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3$

$2AlP + 3H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 2PH_3$

It is poisonous colourless gas contains highly inflammable P_2H_4



bond angle: 93°
Pyramidal

Reactions of Phosphine:

- $PH_3 \xrightarrow{Na} H_2 + Na_3P$
- $PH_3 \xrightarrow{HCl} PH_4Cl$
- $PH_3 \xrightarrow{AgNO_3} Ag_3P + HNO_3$
- $PH_3 \xrightarrow{CuSO_4} Cu_3P_2 + H_2SO_4$
- $PH_3 \xrightarrow{Cl_2} HCl + PCl_5$
- $PH_3 \xrightarrow{N_2O} H_3PO_4 + N_2$
- $PH_3 \xrightarrow{O_2} H_2O + P_2O_5$
- $PH_3 \xrightarrow{HNO_3} P_2O_5 + NO_2$

Phosphorus Trioxide (P_4O_6):

$P_4 + 3O_2 \longrightarrow P_4O_6$ *Acidic Oxide*

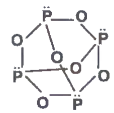
Waxy solid with Garlic odour

$P_4O_6 + 2O_2 \longrightarrow P_4O_{10}$

$P_4O_6 + 4Cl_2 \xrightarrow{\Delta} 2POCl_3 + 2PO_2Cl$

$P_4O_6 + 6H_2O \longrightarrow 4H_3PO_3$

It is anhydride of Phosphorous Acid



P_4O_6

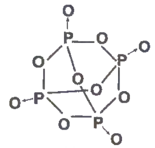
Phosphorous Pentoxide (P_4O_{10}):

$P_4 + 5O_2 \longrightarrow P_4O_{10}$

$P_4O_{10} \xrightarrow{2H_2O} 4HPO_3 \xrightarrow{2H_2O} 2H_4P_2O_7 \xrightarrow{2H_2O} 4H_3PO_4$

It is anhydride of Phosphoric Acid

It is a good drying (dehydrating) agent.



P_4O_{10}

- $P_4O_{10} \xrightarrow{H_2SO_4} SO_3 + HPO_3$
- $P_4O_{10} \xrightarrow{CH_3CONH_2} CH_3CN + HPO_3$
- $P_4O_{10} \xrightarrow{HNO_3} N_2O_5 + HPO_3$
- $P_4O_{10} \xrightarrow{CH_3COOH} (CH_3CO)_2O + HPO_3$

Ammonia:

$N_2 + 3H_2 \underset{Mo}{\overset{Fe}{\longrightarrow}} 2NH_3 \quad \Delta H = -24\ kcal$ ***Habers Method***

$CaNCN + 3H_2O \longrightarrow 2NH_3 + CaCO_3$ ***Cynamide Method***

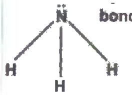
Properties:

- $NH_3 \xrightarrow{Na} H_2 + NaNH_2$
- $NH_3 \xrightarrow{HCl} NH_4Cl$
- $NH_3 \xrightarrow{AgCl} [Ag(NH_3)_2]Cl$
- $NH_3 \xrightarrow{CuO} Cu + N_2 + H_2O$
- $NH_3 \xrightarrow{CaOCl_2} H_2O + N_2 + CaCl_2$
- $NH_3 \xrightarrow{\text{Excess } Cl_2} NCl_3 + HCl$
- $NH_3 \xrightarrow{O_2/Pt} NO + H_2O$
- $NH_3 \xrightarrow{I_2} NH_3.NI_3 + HI$
- $NH_3 \xrightarrow{H_2O_2} H_2O + HNO_2$
- $NH_3 \xrightarrow{K_2HgI_4} H_2NHgO.HgI$

Reactions of Aqueous Ammonia:

- $NH_4OH \xrightarrow{FeCl_3} Fe(OH)_3$
- $NH_4OH \xrightarrow{CuSO_4} Cu(OH)_2 \xrightarrow{(NH_4)_2SO_4} [Cu(NH_3)_4]SO_4$
- $NH_4OH \xrightarrow{CdSO_4} [Cd(NH_3)_4]SO_4$
- $NH_4OH \xrightarrow{AgNO_3} AgOH \xrightarrow{NH_4OH} [Ag(NH_3)_2]OH$
- $NH_4OH \xrightarrow{AlCl_3} Al(OH)_3$ *(white-gelatinum ppt)*
- $NH_4OH \xrightarrow{ZnSO_4} Zn(OH)_2 \xrightarrow{(NH_4)_2SO_4} [Zn(NH_3)_4]SO_4$

Structure : Pyramidal



bond angle: 107°

R. K. MALIK'S NEWTON CLASSES, RANCHI

Characteristics

O, S, Se, Te, Po are called **Chalcogens**

ns^2, np^4. Common Oxidation States are -2, +2, +4, +6

Atomic Radii and Density order: $O < S < Se < Te$

Ionisation Energy and Electronegativity order: $O > S > Se > Te$

O_2 (O=O) is diatomic but others are octaatomic (S_8)

Acidic nature of hydrides: $H_2Te > H_2Se > H_2S > H_2O$

Reducing nature: $H_2Te > H_2Se > H_2S$

Stability Order: $H_2O > H_2S > H_2Se > H_2Te$

Boiling Points Order: $H_2O > H_2Te > H_2Se > H_2S$

Allotropic forms of S are **Rhombic (Octahedral), Monoclinic (Prismatic)**, Plastic etc.

Sulphur (R) $\underset{95.6}{\rightleftharpoons}$ Sulphur (M)

Ozone (O_3)

$3O_2 \underset{\text{Electrical Discharge}}{\overset{\text{Silent}}{\rightleftharpoons}} 2O_3 \quad \Delta H = +68$ KCals

$3F_2 + 3H_2O \longrightarrow 6HF + O_3$

Unstable, diamagnetic, pale blue gas

$2O_3 \xrightarrow{\Delta} 3O_2$

MnO_2, Ag, Pt or PbO_2 are catalysts

It is a powerful oxidising agent

$O_3 \longrightarrow O_2 + (O)$

reactant	product
$K_4[Fe(CN)_6]$	$K_3[Fe(CN)_6]$
PbS	$PbSO_4$
I_2/H_2O	HIO_3
$FeSO_4/H^+$	$Fe_2(SO_4)_3$
KI	I_2
K_2MnO_4	$KMnO_4$
$NaNO_2$	$NaNO_3$
Ag	Ag_2O
Ag_2O	Ag
HCl	Cl_2
S	H_2SO_4
Hg	Hg_2O

denotes reaction with Ozone

$3SO_2 + O_3 \longrightarrow 3SO_3$

$3SnCl_2 + 6HCl + O_3 \longrightarrow 3SnCl_4 + H_2O$

All the three O atoms get used up in oxidation

$CH_2{=}CH_2 + O_3 \xrightarrow{H_2O/Zn} HCHO$

$H_2O_2 + O_3 \longrightarrow H_2O + 2O_2$

$BaO_2 + O_3 \longrightarrow BaO + 2O_2$

Mutual Reduction

Structure

Hydrogen Sulphide

$FeS + H_2SO_4$ (dil) $\longrightarrow H_2S + FeSO_4$

$Sb_2S_3 + 6HCl \longrightarrow 3H_2S + 2SbCl_3$

Colourless gas with rotten egg smell

$2H_2S + 3O_2 \xrightarrow{\Delta} 2H_2O + 2SO_2$

Weak dibasic acid and forms NaHS and Na_2S salts

In acidic medium II group cations (Zn^{2+}, Ni^{2+}, Mn^{2+}, Fe^{2+}) are precipitated as their sulphides.

It is a strong reducing agent

Structure

reactant	product
$Cr_2O_7^{2-}$	Cr^{3+}
MnO_4^-	Mn^{2+}
O_3	O_2
HNO_3	NO_2
H_2O_2	H_2O
SO_2	S
Cl_2	HCl
H_2SO_4	SO_2

denotes reaction with H_2S

Hydrogen Peroxide

$BaO_2 + H_2SO_4 \longrightarrow BaSO_4 + H_2O_2$

Electrolysis of 50% H_2SO_4 or NH_4HSO_4

$H_2SO_4 \rightleftharpoons H^+ + H_2O_4^-$

At Anode: $2HSO_4^- \longrightarrow H_2S_2O_8 + 2e^-$

$H_2S_2O_8 + 2H_2O \longrightarrow H_2O_2 + 2H_2SO_4$

At Cathode: $2H^+ + 2e^- \longrightarrow H_2$

2-Ethyl anthraquinol $\underset{H_2/Pd}{\overset{O_2}{\rightleftharpoons}} H_2O_2$ + 2-Ethyl anthraquinone

Unstable and decomposes $2H_2O_2 \rightleftharpoons 2H_2O + O_2$

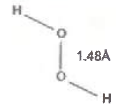
+ve catalysts are MnO_2, C, alkali, Pt, etc; -ve catalysts Acetanilide, alcohol, acid etc

30% H_2O_2 solution is 100 volumes (100v) Perhydrol

Molarity of H_2O_2 = volume strength/11.2

It is an oxidising, bleaching and reducing agent

Structure



reactant	product
PbS	$PbSO_4$
$NaNO_2$	$NaNO_3$
I^-	I_2
H_2S	S
Mn^{2+}/OH^-	MnO_2
Fe^{2+}	Fe^{3+}
HCHO	$HCOOH + H_2$
$Cr_2O_7^{2-}/H^+$	CrO_5
Cr^{3+}/OH^-	CrO_4^{2-}
Hg	HgO
O_3	O_2
Ag_2O	Ag
MnO_2	$MnSO_4$
PbO_2	PbO
Cl_2	HCl
MnO_4^-/H^+	Mn^{2+}
NaOCl	NaCl
KIO_4	KIO_3
$K_3Fe(CN)_6/OH^-$	$K_4[Fe(CN)_6]$
Ce^{4+}	Ce^{3+}

denotes reaction with H_2O_2

Oxidising Agent

Reducing Agent

Oxides of Sulphur

Sulphur Dioxide (SO_2)

$S + 2H_2SO_4$ *(conc)* $\longrightarrow 3SO_2 + 2H_2O$

$Cu + 2H_2SO_4$ *(conc)* $\longrightarrow CuSO_4 + 2H_2O + SO_2$

$2ZnS + 3O_2 \xrightarrow{\Delta} 2ZnO + 2SO_2$

Colourless gas with a pungent odour

$SO_2 + H_2O \longrightarrow H_2SO_3$, *hence anhydride of H_2SO_3*

It is a reducing agent and bleaches due to reduction

$Cl_2 \longrightarrow HCl$; $MnO_4^- \longrightarrow Mn^{2+}$; $IO_3^- \longrightarrow I_2$; $Fe^{3+} \longrightarrow Fe^{2+}$; $Cr_2O_7^{2-} \longrightarrow Cr^{3+}$

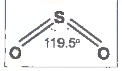
With stronger reducing agents, it acts as an oxidising agent

$H_2S \longrightarrow S$; $CO \longrightarrow CO_2$; $H_2 \longrightarrow H_2O$; $Sn^{2+} \longrightarrow Sn^{4+}$; $Mg \longrightarrow Mg^{2+}$

It bleaches in the presence of moisture due to liberation of nascent H

$SO_2 + 2H_2O \longrightarrow H_2SO_4 + 2[H]$

Structure



Sulphur Trioxide (SO_3)

$2SO_2 + O_2 \overset{Pt}{\rightleftharpoons} 3SO_3$

$6H_2SO_4 + P_4O_{10} \longrightarrow 4H_3PO_4 + 6SO_3$

$Fe_2(SO_4)_3 \xrightarrow{\Delta} Fe_2O_3 + 3SO_3$

Acidic Nature $SO_3 > SO_2$

$SO_3 + H_2O \longrightarrow H_2SO_4$ *hence anhydride of H_2SO_4*

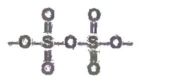
$SO_3 + CaO \longrightarrow CaSO_4$

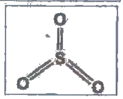
$SO_3 + H_2SO_4 \longrightarrow H_2S_2O_7$ *oleum*

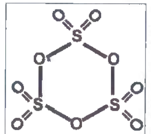
SO_3 is an oxidising agent. $2SO_3 + S \longrightarrow 3SO_2$

$P \longrightarrow P_2O_5$; $HBr \longrightarrow Br_2$; $PCl_3 \longrightarrow Cl_2 + POCl_3$

Structure







R. K. MALIK'S NEWTON CLASSES, RANCHI

Structures of oxoacids of Sulphur

I Sulphurous Acid series:

- H_2SO_3 - Sulphurous Acid — HO–S(=O)–OH S(+IV)
- $H_2S_2O_5$ - di- or Pyrosulphurous Acid — HO–S(=O)(=O)–S(=O)–OH S(+V), S(+III)
- $H_2S_2O_4$ - dithionous acid — HO–S(=O)–S(=O)–OH S(+III), S(+III)

II Sulphuric Acid series:

- H_2SO_4 - Sulphuric Acid — HO–S(=O)(=O)–OH S(+VI)
- $H_2S_2O_3$ - Thiosulphuric Acid — HO–S(=O)(=S)–OH S(+VI), S(-II)
- $H_2S_2O_7$ - di or pyrosulphuric acid (oleum) — HO–S(=O)(=O)–O–S(=O)(=O)–OH S(+VI)

III Thionic Acid series:

- $H_2S_2O_6$ - dithionic acid — HO–S(=O)(=O)–S(=O)(=O)–OH S(+V)
- $H_2S_nO_6$ - Polythionic Acid (n=1to12) — HO–S(=O)(=O)–(S)$_n$–S(=O)(=O)–OH S(+V), S(O)

IV Peroxoacid series:

- H_2SO_5 - peroxomonosulphuric acid (caro's acid) — HO–S(=O)(=O)–O–OH S(+VI)
- $H_2S_2O_8$ - peroxodisulphuric acid (marshall's acid) — HO–S(=O)(=O)–O–O–S(=O)(=O)–OH S(+VI)

Oxyacids of Sulphur

Sulphuric Acid (H_2SO_4)

Lead Chamber Process

$2NO\ (Catalyst) + O_2 \longrightarrow 2NO_2$

$2SO_2 + 2NO_2 \longrightarrow 2SO_3 + 2NO$

$SO_3 + H_2O \longrightarrow H_2SO_4$

Contact Process

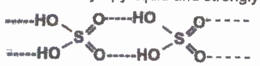
$S + O_2 \xrightarrow{\Delta} SO_2$

$2SO_2 + O_2 \xrightarrow{V_2O_5} 2SO_3$

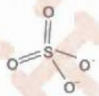
$SO_3 + H_2O \longrightarrow H_2SO_4 \xrightarrow{SO_3} H_2S_2O_7$

$H_2S_2O_7 + H_2O \longrightarrow 2H_2SO_4$

Colourless syrupy liquid and strongly Hydrogen bonded



It is a strong dibasic acid , oxidising agent and dehydrating agent



Tetrahedral

sp³ hybridised

reactant	product
$C_{12}H_{22}O_{11}$	12C
S	SO_2
CaF_2	$CaSO_4$ + HF
KBr	$SO_2 + Br_2$
FeS	$FeSO_4 + H_2S$
K_2MnO_4	$KMnO_4$
Cu	$CuSO_4 + SO_2$
NaOH	$NaHSO_4$, Na_2SO_4
KI	$SO_2 + I_2$
PCl_5	SO_2Cl_2
P	$H_3PO_4 + SO_2$

reactant / product

denotes reaction with H_2SO_4

Sulphurous Acid (H_2SO_3)

$2SO_2 + H_2O \longrightarrow 3SO_2$

Dibasic acid and forms $NaHSO_3$ and Na_2SO_3 salts

It is a reducing and bleaching agent

The O.S of Sulphur is +4

Structure

Pyramidal

sp³ hybridised

Sodium Thiosulphate ($Na_2S_2O_3.5H_2O$)

$Na_2SO_3 + S \longrightarrow Na_2S_2O_3$

$Na_2S + I_2 + Na_2SO_3 \longrightarrow Na_2S_2O_3 + 2Na_2I$ *(springs reaction)*

$4S + 6NaOH \longrightarrow Na_2S_2O_3 + 2Na_2S + 3H_2O$

Colourless, crystalline and efflorescent substance

reactant	product
HCl	SO_2 + S
I_2	$NaI + Na_2S_4O_6$
Cl_2	HCl + S
$CuSO_4$	$Cu_2S_2O_3$, $Na_4[Cu6(S_2O_3)_5]$
AgBr	$Na_3[Ag(S_2O_3)_2]$ + NaBr
$AgNO_3$	$Ag_2S_2O_3$, Ag_2S
$FeCl_3$	$FeCl_2 + Na_2S_4O_6$

reactant / product

denotes reaction with $Na_2S_2O_3$

Halogens (VII A)

Characteristics

- F, Cl, Br and At are the members of this group
- ns^2np^5. Common oxidation states are -1, +1, +3, +5 and +7. +4 and +6 oxidation states are possible in oxides, oxy-acids. F exhibits only -1 oxidation state.
- Atomic radii, Ionic radii and density increase from F to I.
- Electronegativity order is F>Cl>Br>I
- Electron Affinity order is Cl>F>Br>I
- Oxidising nature order is F>Cl>Br>I
- Reducing nature order is $Cl^-<Br^-<I^-$
- Bond Dissociation Energy order Cl–Cl>Br–Br>F–F>I–I
- Reactivity order of non-metals $X_2>H_2>O_2>N_2$

Interhalogen Compounds

Ax Type: ClF, BrF, BrCl, ICl, IBr

More polar the A–X bond, greater the thermal stablity

IF > BrF > ClF > ICl > IBr > BrCl

Linear Sp^3 **Hybridisation**

Ax_3 Type: ClF_3, BrF_3, ICl_3, IF_3

T-Shaped, Central atom is Sp^3d **Hybridised**

Ax_5 Type: ClF_5, BrF_5, IF_5

Square Pyramidal, Sp^3d^2 **Hybridisation**

Ax_7 Type: IF_7

Pentagonal bipyramidal, Sp^3d^3 **Hybridisation**

Hydrolysis: Smaller halogen forms halides, while bigger halogen forms oxyhalides

$ICl + H_2O \longrightarrow HCl + HOI$

$ICl_3 + 2H_2O \longrightarrow 3HCl + HIO_2$

$BrF_5 + 3H_2O \longrightarrow 5HF + HBrO_3$

$IF_7 + 6H_2O \longrightarrow 7HF + H_5IO_6$

ICl is called Wij's reagent and is used in the estimation of Iodine number of Fats and Oils

Bleaching Powder ($CaOCl_2.H_2O$)

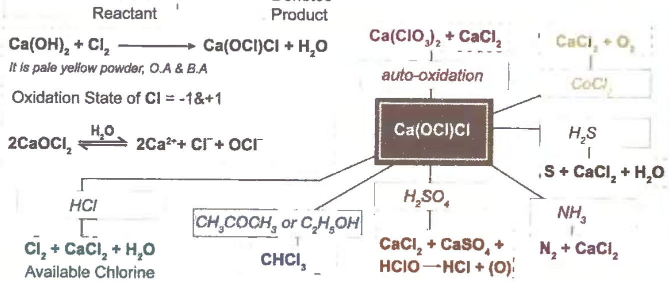
Denotes Reactant | Denotes Product

$Ca(OH)_2 + Cl_2 \longrightarrow Ca(OCl)Cl + H_2O$

It is pale yellow powder, O.A & B.A

Oxidation State of Cl = -1&+1

$2CaOCl_2 \xrightleftharpoons{H_2O} 2Ca^{2+} + Cl^- + OCl^-$



Pseudohalogens

CN^-, SCN^-, N_3^-, OCN^-, NCN^{2-} are the **Pseudohalide ions**

Contains two or more electronegative atoms in which one is N-atom

Properties are similar to halide ions

Dimers are called pseudohalogens

$(CN)_2$ - Cyanogen, $(SCN)_2$ - Thiocyanogen etc

Hydrogen Halides (OR) Hydrohalic Acids

HF, HCl, HBr, HI

$2KHF_2 \xrightarrow{\Delta} K_2F_2 + H_2F_2$ *(or)* $CaF_2 + H_2SO_4 \xrightarrow{\Delta} CaSO_4 + H_2F_2$

$H_2 + Cl_2 \longrightarrow 2HCl$ *(or)* $2NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + 2HCl$

$H_2 + Br_2 \xrightarrow{Pt} 2HBr$ *(or)* $3KBr + H_3PO_4 \longrightarrow K_3PO_4 + 3HBr$

$H_2 + I_2 \underset{450°C}{\overset{Pt}{\rightleftharpoons}} 2HI$ *(or)* $3KI + H_3PO_4 \longrightarrow K_3PO_4 + 3HI$

Stability Order HF > HCl > HBr > HI

Acidic Strength HI > HBr > HCl > HF

Reducing Nature HF < HCl < HBr < HI

Boiling Point order HCl < HBr < HI < HF

Dipole Moment order HI < HBr < HCl < HF

Action with $AgNO_3$

$HCl + AgNO_3 \longrightarrow HNO_3 + AgCl$ (white precipitate)

$HBr + AgNO_3 \longrightarrow HNO_3 + AgBr$ (pale yellow precipitate)

$HI + AgNO_3 \longrightarrow HNO_3 + AgI$ (yellow precipitate)

Action with Halogens

$2HCl + F_2 \longrightarrow 2HF + Cl_2$

$2HBr + Cl_2 \longrightarrow 2HCl + Br_2$

$2HI + Br_2 \longrightarrow 2HBr + I_2$

Oxyacids of Chlorine

Hypochlorous Acid (HClO)

$2Cl_2 + 2HgO + H_2O \longrightarrow 2HClO + HgO.HgCl_2$

$2CaOCl_2 + H_2O + CO_2 \longrightarrow 2HClO + CaCl_2 + CaCO_3$

It is unstable and decomposes

$HClO \longrightarrow HCl + (O)$

Hence it is a powerful oxidising agent and bleaching agent

Structure:

Cl—O^- *Linear* Cl = sp^3

Chlorous Acid ($HClO_2$)

$2(BaClO_2)_2 + H_2SO_4 \longrightarrow 2HClO_2 + BaSO_4$

It undergoes auto oxidation

$2HClO_2 \longrightarrow HClO + HClO_3$

Structure:

Angular Cl = sp^3

Chloric Acid ($HClO_3$)

$Ba(ClO_3)_2 + H_2SO_4 \longrightarrow 2HClO_3 + BaSO_4$

$6NaOH + 3Cl_2 \xrightarrow{hot} NaCl + NaClO_3 + 3H_2O$

Structure:

pyramidal Cl = sp^3

Perchloric Acid ($HClO_4$)

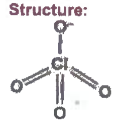
$KClO_4 + H_2SO_4 \longrightarrow 2HClO_4 + KHSO_4$

It is the strongest acid and fumes in moist air

$HClO_4 + Mg \longrightarrow Mg(ClO_4)_2 + H_2$

$Mg(ClO_4)_2$ is called anhydrone, an effective desicant

Structure:



Tetrahedral Cl = sp^3

Acidic Strength: $HClO < HClO_2 < HClO_3 < HClO_4$

Oxidising Strength: $HClO > HClO_2 > HClO_3 > HClO_4$

Oxides of Chlorine

Dichlorine Monoxide (Cl_2O) *Oxidation state= +1*

$2Cl_2 + 2HgO \xrightarrow{0°C} Cl_2O + HgO.HgCl_2$

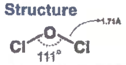
It is anhydride of hypochlorous acid

$2Cl_2 + H_2O \rightleftharpoons 2HClO$ *(Golden Yellow Solution)*

$3Cl_2O + 10NH_3 \longrightarrow 2N_2 + 6NH_4Cl + 3H_2O$

$Cl_2O + 2HCl \longrightarrow 2Cl_2 + H_2O$

Structure



Chlorine Dioxide (ClO_2) *Oxidation state= +4*

$2AgClO_2 + Cl_2 \longrightarrow 2ClO_2 + 2AgCl$

$2NaClO_2 + Cl_2 \longrightarrow 2ClO_2 + 2NaCl$

It is a mixed anhydride of chlorous and chloric acid

$2ClO_2 + H_2O \rightleftharpoons HClO_2 + HClO_3$

$2ClO_2 + 2KOH \longrightarrow 2KClO_2 + KClO_3 + H_2O$

It contains a 3 e⁻ bond and is paramagnetic

Structure

Dichlorine Hexoxide (Cl_2O_6) *Oxidation state=+6*

$2ClO_2 + 2O_3 \xrightarrow{0°C} Cl_2O_6 + 2O_2$

It is a mixed anhydride of chloric and perchloric acids

$2Cl_2O_6 + H_2O \longrightarrow HClO_3 + HClO_4$

$Cl_2O_6 + 2NaOH \longrightarrow NaClO_3 + NaClO_4 + H_2O$

$Cl_2O_6 + HF \longrightarrow HClO_4 + FClO_2$ *(Chloryl Fluoride)*

Cl_2O_6 *(Diamagnetic)* $\rightleftharpoons 2ClO_3$ *(Paramagnetic)*

Dichlorine Heptoxide (Cl_2O_7) *Oxidation state=+7*

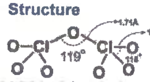
$2HClO_4 + P_2O_5 \longrightarrow Cl_2O_7 + 2HPO_3$

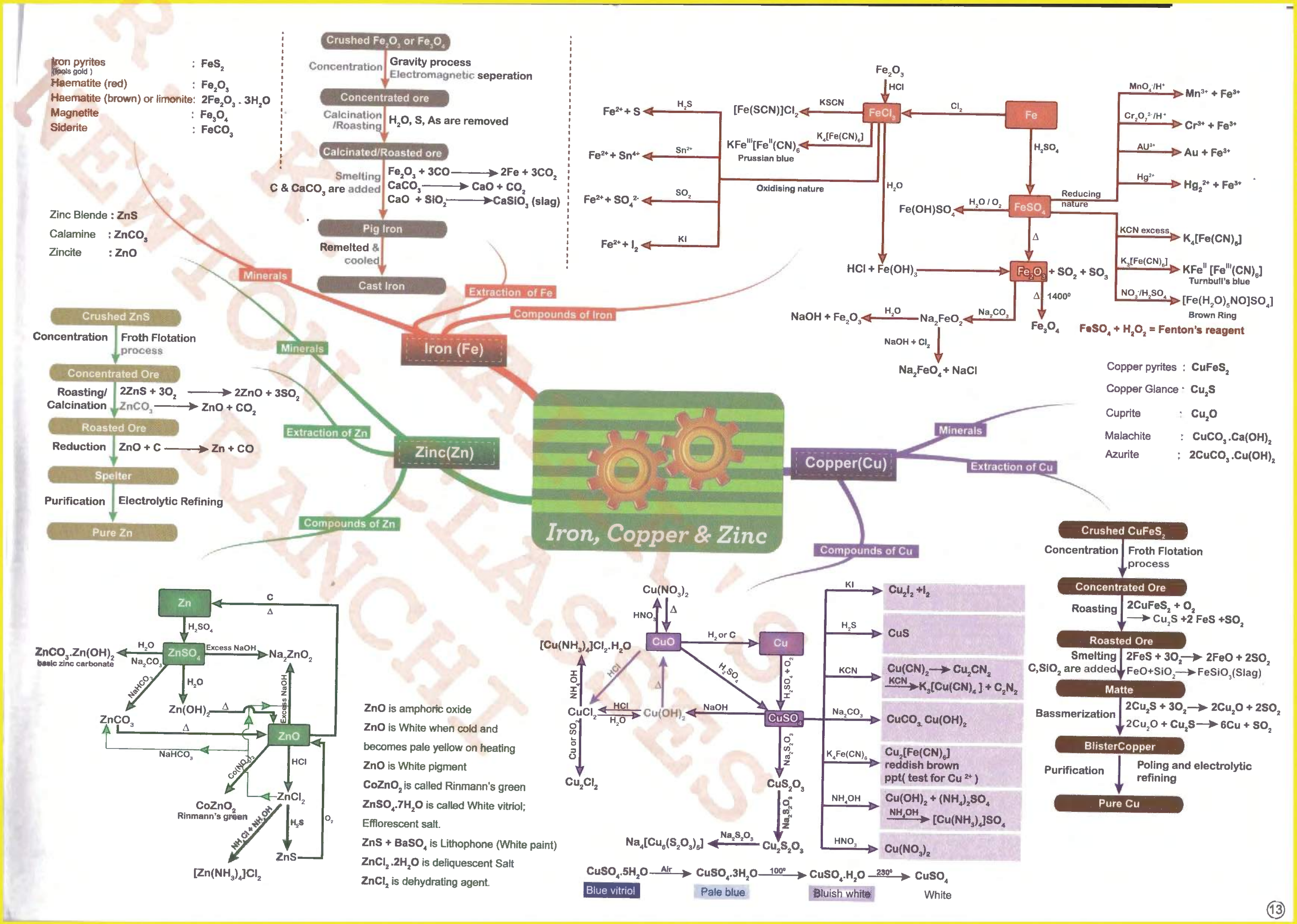
It is an anhydride of perchloric acid

$2Cl_2O_7 + H_2O \longrightarrow 2HClO_4$

$Cl_2O_7 + 2KOH \longrightarrow 2KClO_4 + H_2O$

Structure





R. K. MALIK'S NEWTON CLASSES, RANCHI

Tin, Lead and Silver

Tin(Sn):

Minerals

Cassiterite SnO_2 (Tin Stone)

contains Walframite($FeWO_4$ & $MnWO_4$) as magnetic impurity

Extraction of Sn

Crushed SnO_2 ore

Concentration: Magnetic Separation, Gravity Separation

Concentrated Ore

Roasting: Volatile Impurities SO_2, As_2O_3 are removed

Roasted Ore

Smelting: C & lime stone are added, SnO_2 is reduced, $SnO_2 + C \longrightarrow Sn + CO_2$

Crude Sn

Purification: Liquation, Poling

Pure Zn

Compounds of Sn

$SnCl_4 \xrightarrow{H_2O} Sn(OH)_4 + HCl$

$SnCl_4 \xrightarrow{HCl} H_2SnCl_6$ (chlorostannic acid)

$SnCl_4 \xrightarrow{NH_4Cl} (NH_4)_2SnCl_6$ (Ammonium chlorostannate)

$Sn \xrightarrow{Cl_2} SnCl_4$

$Sn \xrightarrow{HNO_3} H_2SnO_3 + NO_2 + H_2O$

$Sn \xrightarrow{HCl} SnCl_2$

$SnCl_2 \xrightarrow{H_2S} SnS\downarrow \xrightarrow{(NH_4)_2S_2} (NH_4)_2SnS_3$ (Thiostanate)

$SnCl_2 \xrightarrow{NaOH} Sn(OH)_2 \xrightarrow{NaOH} SnO_2$

$Sn(OH)_2 \xrightarrow{CO_2, \Delta} SnO$

$SnO \xrightarrow{HCl} SnCl_2$

$SnO \xrightarrow{NaOH} SnO_2$

$SnO_2 \xrightarrow{O_2} SnO_3$

$SnO \xrightarrow{Na_2CO_3, \Delta}$

$SnO \xrightarrow{O_2} SnO_2$

$SnO_2 \xrightarrow{NaOH} SnO_3$

H_2SnO_3 (Metastannic acid) $\longrightarrow SnO_2$

$SnO_2 \xrightarrow{H_2SO_4} Sn(SO_4)_2$

$SnCl_2 \xrightarrow{HgCl_2} Hg_2Cl_2 \xrightarrow{SnCl_2} Hg + SnCl_4$ (Gray)

$SnCl_2 \xrightarrow{Fe^{3+}} Fe^{2+}$

$SnCl_2 \xrightarrow{Cu^{2+}} Cu^+$

$SnCl_2 \xrightarrow{Au^{3+}} Au$

$SnCl_2 \xrightarrow{I_2/H^+} I^-$

$SnCl_4.5H_2O \longrightarrow$ Butter of Tin

Lead(Pb)

Minerals

Galena : PbS

Anglesite : $PbSO_4$

Cerusite : $PbCO_3$

Lanakite : $PbO.PbSO_4$

Extraction of Pb

Crushed (PbS) galena

Concentration: Froth Floatation process

Concentrated ore

Roasting: $2PbS + 3O_2 \longrightarrow 2PbO + 2SO_2$; $PbS + 2O_2 \longrightarrow PbSO_4$

Roasted ore

Self Reduction: $2PbO + PbS \longrightarrow 3Pb + SO_2$; $PbSO_4 + PbS \longrightarrow 2Pb + 2SO_2$

Crude Pb

Purification: Parkes process, Electrolytic Refining ($PbSiF_6$ Electrolyte)

Pure Pb

Compounds of Pb

$Pb \xrightarrow{Cl_2} PbCl_4$

$Pb \xrightarrow{HCl} PbCl_2$

$PbCl_2 \xrightarrow{H_2O} Pb(OH)Cl$

$PbCl_2 \xrightarrow{KI} PbI_2\downarrow$ (yellow)

$PbCl_2 \xrightarrow{HCl} H_2PbCl_4$

$PbCl_2 \xrightarrow{K_2CrO_4} PbCrO_4\downarrow$ (yellow)

$PbCl_2 \xrightarrow{H_2S} PbS \xrightarrow{O_2} PbO$

$PbO \xrightarrow{HNO_3} O_2 + NO_2 + Pb(NO_3)_2$

$PbO \xrightarrow{C, \Delta} CO + Pb$

$PbO \xrightarrow{NaOH} [Pb(OH)_4]^{2-}$

$PbO \xrightarrow{O_2} Pb_3O_4$

$Pb_3O_4 \xrightarrow{HCl} PbSO_4 + H_2O + O_2$

$Pb_3O_4 \xrightarrow{HCl} PbCl_2$

$Pb_3O_4 \xrightarrow{C, \Delta} Pb$

$Pb_3O_4 \xrightarrow{HNO_3} PbO_2 + Pb(NO_3)_2$

$PbO_2 + Pb(NO_3)_2 \xrightarrow{\Delta} PbO$

$PbO_2 \xrightarrow{H_2SO_4 \text{ or } SO_2} PbSO_4$

$PbO_2 \xrightarrow{HCl} PbCl_2$

$PbO_2 \xrightarrow{NaOH} [Pb(OH)_6]^{2-}$

PbO(Buff coloured) $\longrightarrow$ Litharge

PbO(Yellow) $\longrightarrow$ massicot

Pb_3O_4 ($PbO + PbO_2$) $\longrightarrow$ Red lead

$Pb(CH_3COO)_2 \longrightarrow$ Sugar of Lead

$2PbCO_3.Pb(OH)_2$ is white lead (Basic Lead carbonate)

Silver (Ag)

Minerals

Argentite : Ag_2S (Silver glance)

Pyrargyrite : $3Ag_2S.Sb_2S_3$ (Ruby silver)

Stromeyerite : $(Cu,Ag)_2S$ (Copper glance)

Argentiferous galena : $PbS.Ag_2S$

Chlorargyrite : AgCl (Horn Silver)

Extraction of Ag

"Cyanide or Mac-Arthur and Forest process"

Crushed Argentite (Ag_2S)

Concentration: Froth flotation process

Concentrated Ore

Cyanidation

$Ag_2S + 4NaCN \longrightarrow 2Na[Ag(CN)_2] + Na_2S \xrightarrow[H_2O]{O_2} Na_2SO_4$

$AgCl + 2NaCN \longrightarrow Na[Ag(CN)_2] + NaCl$

Precipitation of Ag

$2Na[Ag(CN)_2] + Zn \longrightarrow 2Ag + Na_2[Zn(CN)_4]$

Impure Ag

Purification: Cupellation, Electrolytic refining

Pure Ag

Compounds of Ag

$Ag \xrightarrow{HNO_3} AgNO_3$

$AgNO_3 \xrightarrow{\Delta} O_2 + AgNO_2$

$AgNO_3 \xrightarrow{\Delta, \text{ red hot}} Ag + NO_2 + O_2$

$AgNO_3 \xrightarrow{HX} NaNO_3 + AgX\downarrow$

$AgNO_3 \xrightarrow{NaOH} Ag_2O + NaNO_3 + H_2O$

$AgNO_3 \xrightarrow{KCN} AgCN\downarrow$ (white) $\xrightarrow{KCN} K[Ag(CN)_2]$ (Soluble)

$AgNO_3 \xrightarrow{Na_2S_2O_3} Ag_2S_2O_3$ (white) $\xrightarrow{H_2O} Ag_2S\downarrow$ (Black) $+ H_2SO_4$

$Ag_2S_2O_3 \xrightarrow[\text{excess}]{Na_2S_2O_3} Na_3[Ag(S_2O_3)_2]$

$AgNO_3 \xrightarrow{I_2, \text{ less}} HNO_3 + AgI + AgIO_3$

$AgNO_3 \xrightarrow[\text{excess}]{I_2} HNO_3 + AgI + HIO_3$

$AgNO_3 \xrightarrow{Cu} Ag + Cu(NO_3)_2$

$AgNO_3 \xrightarrow{Zn} Ag + Zn(NO_3)_2$

$AgNO_3 \xrightarrow{PH_3} HNO_3 + H_3PO_3 + Ag$

$AgNO_3 \xrightarrow{NH_4OH} Ag_2O + NH_4NO_3 + H_2O$

$\xrightarrow{NH_4OH} [Ag(NH_3)_2]NO_3$ (Tollen's Reagent)

$[Ag(NH_3)_2]NO_3 \xrightarrow{C_2H_2} Ag_2C_2$

$[Ag(NH_3)_2]NO_3 \xrightarrow{C_6H_{12}O_6 \text{ (glucose)}} C_6H_{12}O_7$ (gluconic acid) + Ag

$[Ag(NH_3)_2]NO_3 \xrightarrow{CH_3CHO} HCOOH + Ag$

$AgNO_3$ is called Lunar caustic

Acidic Radicals (Anion) Analysis

Action with dil.HCl or dil.H_2SO_4

Carbonate(CO_3^{2-}):

* Brisk Effervescence of colourless and odourless gas (CO_2)

$Na_2CO_3 + H_2SO_4 \rightarrow Na_2CO_3 + H_2O + CO_2\uparrow$

* The gas liberated turns Lime water milky but milkyness disappears on passing the gas in excess

$Ca(OH)_2 + CO_2 \rightarrow \underset{(Milky)}{CaCO_3\downarrow} + H_2O \xrightarrow{CO_2} \underset{(Soluble)}{Ca(HCO_3)_2}$

Sulphide(S^{2-}):

* The salt gives clourless gas with the smell of rotten eggs(H_2S)

$Na_2S + H_2SO_4 \rightarrow Na_2SO_4 + H_2S$

* The gas turns Lead acetate paper black

$H_2S + Pb(CH_3COO)_2 \rightarrow \underset{(Black)}{PbS\downarrow} + 2CH_3COOH$

* $\underset{(Sodium\ Nitroprusside)}{Na_2S + Na_2[Fe(CN)_5NO]} \rightarrow \underset{(purple)}{Na_4[Fe(CN)_5NOS]}$

Sulphite(SO_3^{2-}):

* The salt gives clourless gas with pungent smell of burning sulphur(SO_2)

$Na_2SO_3 + H_2SO_4 \rightarrow SO_2\uparrow + Na_2SO_4 + H_2O$

* The gas turns $K_2Cr_2O_7/H^+$ paper to green.

$K_2Cr_2O_7 + H_2SO_4 + SO_2 \rightarrow \underset{(green)}{Cr_2(SO_4)_3} + K_2SO_4 + H_2O$

* The gas (SO_2) in excess decolourises dil.$KMnO_4$ Solution

$\underset{Pink}{2KMnO_4} + 2H_2O + 3SO_2 \rightarrow \underset{(Colourless)}{3MnSO_4} + K_2SO_4 + 2H_2SO_4$

Action with $BaCl_2$

Sulphate(SO_4^{2-}):

* Na_2CO_3 extract +dil HCl + $BaCl_2$ solution

$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4\downarrow + 2NaCl$

* The percipitate is insoluble in conc.HNO_3.

Action with Conc. H_2SO_4

Nitrate (NO_3^-):

* The salt gives pungent brown fumes (NO_2)

$NaNO_3 + H_2SO_4 \rightarrow NaHSO_4 + HNO_3$

$4HNO_3 \rightarrow 4NO_2\uparrow + O_2\uparrow + 2H_2O$

* Thick brown fumes in presence of Cu turnings.

$Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2H_2O + 2NO_2\uparrow$

Brown Ring Test:

$2NaNO_3 + H_2SO_4 \rightarrow 2NaHSO_4 + 2HNO_3$

$2HNO_3 + 6FeSO_4 + 3H_2SO_4 \rightarrow 3Fe_2(SO_4)_3 + 2NO + 4H_2O$

$[Fe(H_2O)_6]SO_4 + NO \rightarrow [Fe(H_2O)_5NO]SO_4 + H_2O$

* Due to charge transfer the complex contains Fe(+1) and NO^+
* Magnetic moment = 3.87 B.M *(3 unpaired electrons)*.

Chloride (Cl^-):

* The salt gives colourless gas with pungent smell (HCl).

$NaCl + H_2SO_4 \rightarrow NaSO_4 + HCl(\uparrow)$

* HCl obtained gives the following tests

$HCl + NH_4OH \rightarrow \underset{(White\ fumes)}{NH_4Cl + H_2O}$

$4HCl + MnO_2 \rightarrow MnCl_2 + 2H_2O + \underset{(Greenish\ yellow\ gas)}{Cl_2(\uparrow)}$

$HCl + AgNO_3 \rightarrow \underset{(White)}{AgCl(\downarrow)} + HNO_3$

$AgCl + 2NH_4OH \rightarrow \underset{(Soluble)}{[Ag(NH_3)_2]Cl} + H_2O$

Chromyl Chloride Test:-

$NaCl + H_2SO_4 \xrightarrow{\Delta} NaHSO_4 + HCl$

$K_2Cr_2O_7 + H_2SO_4 \xrightarrow{\Delta} 2NaHSO_4 + CrO_3 + H_2O$

$CrO_3 + 2HCl \rightarrow \underset{(Reddish\ brown\ vapours)}{CrO_2Cl_2 + H_2O}$

$CrO_2Cl_2 + 4NaOH \rightarrow \underset{(Yellow\ Solution)}{Na_2CrO_4} + 2NaCl + 2H_2O$

$Na_2CrO_4 + Pb(CH_3COO)_2 \rightarrow 2CH_3COONa + \underset{(Yellow)}{PbCrO_4(\downarrow)}$

Bromide (Br^-):

* The salt gives brown fumes of Br_2

$NaBr + H_2SO_4 \rightarrow NaHSO_4 + HBr$

$2HBr + H_2SO_4 \rightarrow Br_2\uparrow + SO_2 + 2H_2O$

* Thick brown fumes are evolved with MnO_2

$2NaBr + 3H_2SO_4 + MnO_2 \rightarrow 2NaHSO_4 + 2MnSO_4 + 2H_2O + Br_2$

$NaBr + AgNO_3 \rightarrow \underset{(Pale\ Yellow)}{AgBr(\downarrow)} + NaNO_3$

AgBr is partially soluble in NH_4OH

$AgBr + 2NH_4OH \rightarrow [Ag(NH_3)_2]Br + H_2O$

Iodide(I^-):

* The salt gives violet fumes(I_2).

$KI + H_2SO_4 \xrightarrow{\Delta} KHSO_4 + HI$

$2HI + H_2SO_4 \rightarrow I_2(\uparrow) + 2H_2O + SO_2$

I_2 + Starch paper → Deep blue complex

$KI + AgNO_3 \rightarrow \underset{(Yellow)}{AgI(\downarrow)} + KNO_3$

* AgI is insoluable in NH_4OH.

Separation of Basic Radicals (Cations) into different groups based on their selective precipitation

Group	Group Reagent	Basic Radical	Colour&Composition
I	Dil. HCl	Ag^+	AgCl, white
		Pb^{2+}	$PbCl_2$, white
II	H_2S + dil.HCl	Hg^{2+}	HgS, black
		Pb^{2+}	PbS, black
		Cu^{2+}	CuS, black
		Bi^{3+}	Bi_2S_3, black
III	$NH_4OH + NH_4Cl$	Al^{3+}	$Al(OH)_3$, white
		Fe^{3+}	$Fe(OH)_3$, reddish brown
		Cr^{3+}	$Cr(OH)_3$, green
IV	$(NH_4)_2S + NH_4OH$	Zn^{2+}	ZnS, greenish white
		Mn^{2+}	MnS, buff
V	$(NH_4)_2CO_3 + NH_4OH$	Ba^{2+}	$BaCO_3$, white
		Ca^{2+}	$CaCO_3$, white
VI	Na_2HPO_4	Mg^{2+}	$MgNH_4PO_4$, white
zero	NaOH	NH_4^+	NH_3 gas evolved

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Basic Radicals (Cation) Analysis

Group Seperation

Zero Group

Group I

Group II

Group III

Group IV

Group V

Group VI

Group I

Ag^+: AgCl dissolves in NH_4OH

$AgCl + 2NH_4OH \longrightarrow [Ag(NH_3)_2]Cl + 2NH_4OH$

$2AgCl(\downarrow) + 2NH_4NO_3 \xleftarrow{2HNO_3}$; $\xrightarrow{2KI} AgI(\downarrow) + KCl + 2NH_3$

white ; yellow

$\xrightarrow{H_2S} Ag_2S + 2HCl + 4NH_3$

$NH_3 + KCl + AgCrO_4(\downarrow) \xleftarrow{K_2CrO_4}$

yellow

Pb^{2+}: $PbCl_2 + 2KI \longrightarrow PbI_2(\downarrow) + 2KCl$

yellow

$PbCl_2 + K_2CrO_4 \longrightarrow PbCrO_4(\downarrow) + 2KCl$

yellow

Zero Group

NH_4^+: Salt + NaOH $\xrightarrow{\Delta}$ NH_3 *gas evolves*

$NH_4Cl + NaOH \longrightarrow NH_3(\uparrow) + NaCl + H_2O$

NH_3 gas gives white fumes with HCl

$NH_3 + HCl \longrightarrow NH_4Cl$ *(white fumes)*

Nessler's Reagent test:

$NH_4Cl + 4KOH + 2K_2HgI_4 \longrightarrow Hg(NH_2)$–O–$Hg(I)\downarrow + 7KI + KCl + 3H_2O$

Brown

Iodide of million's base

Group VI

Mg^{2+}: $MgCl_2 + Na_2HPO_4 + NH_4OH \longrightarrow Mg(NH_4)PO_4(\downarrow) + 2NaCl + H_2O$

white

Group V

Ba^{2+}: $BaCO_3$ dissolves in CH_3COOH.

$BaCO_3 + 2CH_3COOH \longrightarrow Ba(CH_3COO)_2 + H_2O + CO_2$

$\downarrow K_2CrO_4$

$BaCrO_4(\downarrow) + CH_3COOH$

(yellow)

Ca^{2+}: $CaCO_3 + 2CH_3COOH \longrightarrow Ca(CH_3COO)_2 + H_2O + CO_2$

$\downarrow (NH_4)_2C_2O_4$

$CaC_2O_4(\downarrow) + CH_3COONH_4$

(white)

Group II

Hg^{2+}: HgS dissolves in aqua-regia to form $HgCl_2$

$3HgS + 2(HNO_3 + 3HCl) \longrightarrow 3HgCl_2 + 2NO + 3S + 4H_2O$

aqua regia

$2Hg + SnCl_4 \xleftarrow{SnCl_2}$; $\xrightarrow{2KI} HgI_2(\downarrow) + 2KCl$

grey ; orange

$\xrightarrow{KI} K_2HgI_4$ (Soluble)

Nessler's Reagent

Pb^{2+}: PbS dissolves in HNO_3 to form $Pb(NO_3)_2$

$3PbS + 8HNO_3 \longrightarrow 3Pb(NO_3)_2 + 2NO + 3S + 4H_2O$

$PbI_2(\downarrow) \xleftarrow{KI}$; $\xrightarrow[CH_3COOH]{K_2CrO_4} PbCrO_4(\downarrow)$

yellow ; yellow

Cu^{2+}: CuS dissolves in HNO_3 to form $Cu(NO_3)_2$

$3CuS + 8HNO_3 \longrightarrow 3Cu(NO_3)_2 + 2NO + 3S + 4H_2O$

$\downarrow CH_3COOH$

$Cu_2[Fe(CN)_6](\downarrow) \xleftarrow{K_4[Fe(CN)_6]} (CH_3COO)_2Cu + HNO_3$

chocolate brown

Bi^{3+}: Bi_2S_3 dissolves in HNO_3 to form $Bi(NO_3)_3$

$Bi_2S_3 + 8HNO_3 \longrightarrow 2Bi(NO_3)_3 + 2NO + 3S + 4H_2O$

$\downarrow NH_4OH$

$2Bi(OH)_3 + NH_4NO_3$

$\downarrow HCl$

$H_2O + NaCl + Na_2SnO_3 + Bi(\downarrow) \xleftarrow[NaOH]{Na_2SnO_2} BiCl_3$

Group III

Al^{3+}: Gelatinous white ppt of $Al(OH)_3$ dissolves in NaOH

$Al(OH)_3 + NaOH \longrightarrow NaAlO_2 + 2H_2O$

$\downarrow \Delta\ NH_4Cl$

$Al(OH)_3 + NaCl + NH_3$

Fe^{3+}: $Fe(OH)_3$ dissolves in HCl to form $FeCl_3$

$Fe(OH)_3 + 3HCl \longrightarrow FeCl_3 + 3H_2O$

$KCl + Fe_4[Fe(CN)_6]_3 \xleftarrow{K_4[Fe(CN)_6]}$; $\xrightarrow{KSCN} [Fe(SCN)]Cl_2 + KCl$

Prussian Blue (Ferric Ferrocyanide) ; Blood Red

Cr^{3+}: $Cr(OH)_3$ dissolves in NaOH in presence of Br_2 to form Na_2CrO_4

$Cr(OH)_3 \xrightarrow{NaOH + Br_2} Na_2CrO_4$

$\downarrow Pb(CH_3COO)_2$

$PbCrO_4(\downarrow) + CH_3COONa$

yellow

Group IV

Zn^{2+}: ZnS dissolves in dil. HCl to form $ZnCl_2$

$ZnS + 2HCl \longrightarrow ZnCl_2 + H_2S$

$\downarrow \Delta\ NaOH$

$Zn(OH)_2 \xrightarrow{NaOH} Na_2ZnO_2 \xrightarrow{H_2S} ZnS(\downarrow) + NaOH$

Mn^{2+}: MnS dissolves in HCl to form $MnCl_2$

$MnS + 2HCl \longrightarrow MnCl_2 + H_2S$

$MnCl_2 + 2NaOH \longrightarrow Mn(OH)_2 + 2NaCl$

$\downarrow (O)$ in air

$MnO_2 + H_2O$

$MnO_2 + 2HNO_3 \longrightarrow Mn(NO_3)_2 + H_2O + 1/2O_2$

$\downarrow PbO_2 + HNO_3$

$HMnO_4 + Pb(NO_3)_2 + H_2O$

purple

Permanganic Acid

R. K. MALIK'S NEWTON CLASSES, RANCHI

Co-ordination Compounds

Addition or Molecular Compounds

Addition or Molecular Compounds
- Double salts
  - $KCl.MgCl_2.6H_2O \xrightarrow{H_2O}$ K^+, Mg^{2+}, $2Cl^-$ ions
- Complex or Coordination compounds
  - $K_4[Fe(CN)_6] \xrightarrow{H_2O}$ $4K^+$ & $[Fe(CN)_6]^{4-}$ ions (Complex anion)

Co-ordination Compounds contain complex anions or complex cations or complex cation and anion

$[Co(NH_3)_6]Cl_3 \xrightarrow{H_2O} [Co(NH_3)_6]^{3+} + 3Cl^-$ (complex cation)

$[Pt(NH_3)_4][PtCl_4] \xrightarrow{H_2O} [Pt(NH_3)_4]^{2+} + [PtCl_4]^{2-}$ (complex cation and anion)

$[Ni(CO)_4] \xrightarrow{H_2O}$ No ionisation (Neutral Complex)

$[Co(NH_3)_6]Cl_3$
- Metal Cation
- Ligand
- Co-ordination Number

Naming of Co-ordination Compounds

- $[Cr(NH_3)_3(H_2O)_3]Cl_3$ → Triamminetriaquachromium (III) Chloride
- $K_4[Fe(CN)_6]$ → Potassiumhexacyanoferrate (II)
- $[Pt(CN)_2Cl_2]Cl_2$ → Dichlorobis(Ethylene diammine)platinum (IV) Chloride
- $K_2[HgI_4]$ → Potassiumtetraiodomercurate(II)
- $Fe(CO)_5$ → Pentacarbonyliron (O)
- $[Pt(NH_3)_4][CuCl_4]$ → Tetrammineplatinum(II) tetrachloro cuprate (II)

Effective Atomic Number (EAN)

According to sidgwick, a stable complex is formed if the EAN of the central metal is equal to the atomic number (z) of the next noble gas element.

EAN = Z (Metal) - Oxidation state + 2 x Coordination number

eg: $[Co(NH_3)_6]^{3+}$ EAN = 27-3+(2x6) = 36 (kr)

$Ni(CO)_4$ EAN = 28-0+(2x4) = 36 (kr)

$K_2[PtCl_6]$ EAN = 78-4+(2x6) = 86 (Rn)

Isomerism

Ligands

Ligands (Neutral, +ve, or –ve) are Lewis Bases and form co-ordinate bonds with metal by donating electron pairs

The total no of Ligands indicate the Co ordination number

The Ligands containing one, two, or more e pair donor atoms are called monodentate, bidentate etc.

Monodentate Ligands: $NH_3, H_2O, NO, CO, P_4, Cl^-, Br^-, I^-, OH^-, CN^-, CNS^-$

Bidentate Ligands $C_2O_4^{2-}$, $NH_2CH_2CH_2NH_2$ Ethyl diamine(en)

Hexadentate Ligands $CH_2-N(CH_2COO^-)_2$ | $CH_2-N(CH_2COO^-)_2$ (EDTA)

Ambidentate Ligands :

Contain more than one donor atom

← NO_2 (N - nitro) ← O-N=O (O - Nitro) ← C≡N (Cyano) ← NC (IsoCyano)

Hybridization and Geometries

1) Octahedral complexes

(d^2sp^3 or sp^3d^2 hybridization)

Inner Orbital complex or Low spin complexes → d^2sp^3 hybridization
- a) $[Fe(CN)_6]^{4-}$ unpaired e^- = 0, μ = 0 , Diamagnetic
- b) $[Co(NH_3)_6]^{3+}$ unpaired e^- = 0, μ = 0 , Diamagnetic
- c) $[Cr(NH_3)_6]^{3+}$ unpaired e^- = 3, μ = 3.87 BM, Paramagnetic

Outer Orbital complex or High spin complexes → sp^3d^2 hybridization
- a) $[CoF_6]^{3-}$ unpaired e^- = 4, μ = 4.90 BM, Paramagnetic
- b) $[Cu(NH_3)_6]^{2+}$ unpaired e^- = 1, μ = 1.73 BM, Paramagnetic
- c) $[Fe(H_2O)_6]^{3+}$ unpaired e^- = 4, μ = 4.9 BM, Paramagnetic

2) Square Planar Complexes:

dsp^2 hybridization

a) $[Ni(CN)_4]^{2-}$:

	3d	4S	4P
Ni^{2+} = (Z=28)	⇅ ⇅ ⇅ ↑ ↑		

since μ_s = 0 & diamagnetic

	3d	4S	4P
Ni^{2+} = (Rearranged)	⇅ ⇅ ⇅ ⇅ ☐	☐	☐ ☐ ☐

dsp^2 hybridization.

b) $[Cu(NH_3)_4]^{2+}$, unpaired e^- =1, μ = 1.73 BM, Paramagnetic

	3d	4S	4P
Cu^{2+} = (Rearranged)	⇅ ⇅ ⇅ ⇅ ☐	☐	☐ ☐ ↑

Square Planar ; dsp^2 hybridization.

3) Tetrahedral complexes:

sp^3 hybridization
- a) $[Zn(NH_3)_4]^{2+}$ unpaired e^- = 0, μ = 0, Diamagnetic
- b) $[Ni(CO)_4]$ unpaired e^- = 0, μ = 0, Diamagnetic

4) Linear Complexes:

sp hybridization
- a) $[Ag(NH_3)_2]^{2+}$
- b) $[Ag(CN)_2]^{2-}$

μ = 0 , Diamagnetic

(18)

Stereo Isomerism

Constitutional Isomerism

Ionisation Isomerism:
Provide different ions in their aqueous solution

$[Co(NH_3)_5Br]SO_4 \rightarrow [Co(NH_3)_5Br]^{2+} + SO_4^{2-}$
Red Violet

$[Co(NH_3)_5SO_4]Br \rightarrow [Co(NH_3)_5SO_4]^{+} + Br^{-}$
Red

$[Pt(NH_3)_4Cl_2]Br_2$ and $[Pt(NH_3)_4Br_2]Cl_2$

Hydrate Isomerism:
Number of water molecules present inside and outside the co-ordination shpere are different.

$[Cr(H_2O)_6]Cl_3$. $[Cr(H_2O)_5Cl]Cl_2.H_2O$ and $[Cr(H_2O)_4Cl_2]Cl.2H_2O$

$[CoCl(H_2O)(NH_3)_4]Cl_2$ and $[CoCl_2(NH_3)_4]H_2OCl$

Linkage Isomerism:
Differ in the donor atom when ambidentate ligands are present

$[Co(NH_3)_5NO_2]Cl_2$ and $[Co(NH_3)_5ONO]Cl_2$

$[Cr(SCN)(H_2O)_5]^{3+}$ and $[Cr(NCS)(H_2O)_5]^{3+}$

Coordination Isomerism:
Due to interchange of Ligands between cationic and anionic complex ions

$[Co(NH_3)_6][Cr(C_2O_4)_3]$ and $[Cr(NH_3)_6][Co(C_2O_4)_3]$

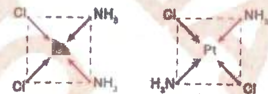
$[Pt(NH_3)_4][CuCl_4]$ and $[Cu(NH_3)_4][PtCl_4]$

Cis-trans isomerism

Common in co-ordination number 4 and 6 compounds
Due to different spatial arrangement of metal-ligand bonds

Co. No. 4 (Square planar complexes)

MA_2B_2 Type: $[Pt(NH_3)_2Cl_2]$

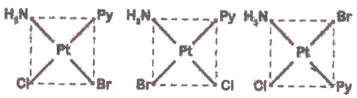


cis — Pt - Cl bonds at 90°
trans — at 180° angle

MA_2BC Type: $[Pt(NH_3)_2ClNO_2]$

cis — trans

MABCD Type: $[Pt(NH_3)PyClBr]$

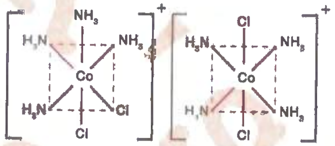


NH_3 - trans to Br — to Cl — to Py

Tetrahedral complexes do not exhibit

Co. No. 6 (Octahedral complexes)

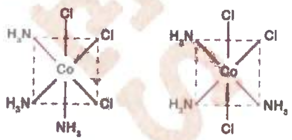
MA_4B_2 Type: $[Co(NH_3)_4Cl_2]^+$



cis(violet) — trans(green) → tree green

MA_6 and MA_5B type do not exhibit.

MA_3B_3 Type: $[Co(NH_3)_3Cl_3]$



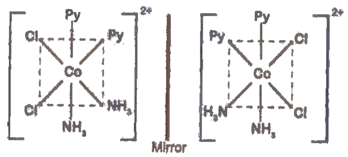
cis(Facial) — trans(Meridonal)

(1,2,3) are cis and (1,2,6) are trans positions

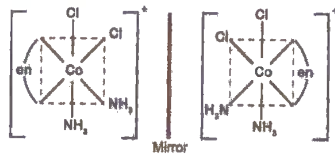
Enantiomerism(optical isomerism)

The molecule should be chiral (no symmetry)
Non identical mirror images of each other are called Eanantiomers
Not found in tetrahedral and square planar complexes
Common in Octahedral complexes

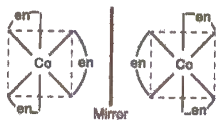
$[MA_2B_2C_2]^{n\pm}$ Type: $[Pt(Py)_2(NH_3)_2Cl_2]^{2+}$



$MAAB_2C_2$ Type: $[Co(en)(NH_3)_2Cl_2]^+$ AA = bidentate ligand



$M(AA)_3$ Type: $[Co(en)_3]^{3+}$



$M(AA)_2B_2$ Type: $[Co(en)_2Cl_2]^+$